



Forecasting government bond yields with large Bayesian vector autoregressions

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ABSTRACT

We propose a new approach to forecasting the term structure of interest rates, which allows to efficiently extract the information contained in a large panel of yields. In particular, we use a large Bayesian Vector Autoregression (BVAR) with an optimal amount of shrinkage towards univariate AR models. The optimal shrinkage is chosen by maximizing the Marginal Likelihood of the model. Focusing on the US, we provide an extensive study on the forecasting performance of the proposed model relative to most of the existing alternative specifications. While most of the existing evidence focuses on statistical measures of forecast accuracy, we also consider alternative measures based on trading schemes and portfolio allocation. We extensively check the robustness of our results, using different datasets and Monte Carlo simulations. We find that the proposed BVAR approach produces competitive forecasts, systematically more accurate than random walk forecasts, even though the gains are small.

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1. Introduction

Producing accurate forecasts of the term structure of interest rates is crucial for bond portfolio management, derivatives pricing, and risk management. Unfortunately, all the forecasting models proposed so far in the macroeconomic and financial literature have a hard time in producing forecasts more accurate than a simple no-change forecast (i.e. a random walk forecast). The existing methods can be roughly categorized in three groups. The first two groups of models have the clear advantage of being grounded on finance theory, while the third group is the one that so far has produced the best results in out of sample forecast accuracy.

The first group contains models based on forward rate regressions. Such models try to forecast the future yields by extracting the information contained in the present forward rates. Prominent examples of this approach are, e.g. [Fama and Bliss \(1987\)](#) and [Cochrane and Piazzesi \(2005\)](#). Even though these papers document the existence of a predictive content in the forward rates, the out of sample forecasts produced by these models are typically outperformed by a simple no-change forecast (see e.g. [Diebold and Li, 2006](#)).

The second group contains models based on the No-Arbitrage paradigm. Typically the practical implementation of these models involves imposing an affine specification on a set of latent factors.

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Affine term structure models perform extremely well in fitting the yield curve in sample (see e.g. [De Jong, 2000](#) and [Dai and Singleton, 2000](#)) but the performance in out of sample forecasting is quite poor. [Duffee \(2002\)](#) has shown that beating a random walk with a traditional no arbitrage affine term structure model is difficult. [Ang and Piazzesi \(2003\)](#) show that imposing no-arbitrage restrictions and an essentially affine specification of market prices of risk improves out-of-sample forecasts from a VAR(12), but the gain with respect to a random walk forecast is small. More favorable evidence in this respect has been found by [Almeida and Vicente \(2008\)](#). In this case one of the reasons for the difference in their results with respect to the rest of the literature ([Duffee \(2011\)](#), [Joslin et al. \(2011\)](#)) is that they also consider models with stochastic volatility while most of the literature only adopts Gaussian models. Also [Favero et al. \(2012\)](#) and [Moench \(2008\)](#) document a rather good performance of the ATSM models, but in both cases such models are complemented with a large macroeconomic dataset.

A third group of papers uses the [Nelson and Siegel \(1987\)](#) exponential components framework ([Diebold and Li, 2006](#)), possibly also imposing on it the no-arbitrage restrictions ([Christensen et al., 2011](#)). The forecasting results obtained by these models are better, with the [Diebold and Li \(2006\)](#) model producing one-year-ahead forecasts that are noticeably more accurate than standard benchmarks. Still, the gains are small at shorter forecast horizons.

In this paper we propose a new strategy for forecasting the term structure of interest rates, which exploits the information

contained in a large panel of government bond yields. Focusing on the US, we show that our proposed strategy produces systematically better forecasts than all the methods outlined above. The method also outperforms the random walk, with small but systematic gains. The starting point of our strategy is the consideration that the yield curve can be thought of as a vector process composed of yields of different maturities. In that light, a straightforward approach to forecast would be to simply run a Vector Autoregression (VAR). However, such a strategy soon encounters the so-called “curse of dimensionality” problem, as the number of parameters to estimate rapidly reduces the degrees of freedom of the VAR system. As a result, the forecasts produced by a VAR are typically poor. To overcome this difficulty, one can either choose to sacrifice completely the cross-sectional information, and estimate e.g. a simple AR model on each yield, or try to summarize the information in an efficient but manageable manner. This latter possibility can be pursued by using a Bayesian Vector Autoregression (BVAR).

A BVAR is a VAR whose coefficients are random variables on which the researcher can impose some a priori information. Without entering into philosophical disputes about the Bayesian and the classical approach in econometrics, we think it is worth stressing here that BVARs can be interpreted simply as a selection device. Consider the first equation of a large VAR: there are many regressors, and the researcher needs to solve the trade-off between using as much information as possible, and the loss in degrees of freedom coming from having too many parameters to estimate. An intuitive way to proceed would be to start with an empty model, then adding a candidate regressor and performing a test of significance for that regressor. If the null is rejected, then the regressor is kept. The procedure can be repeated until the last candidate regressor is considered. What this procedure implicitly does is selecting the regressors on the basis of how much valuable information they contain. Information is valuable if it is able to significantly increase the likelihood of the model. The Bayesian algorithm works similarly. A priori the coefficient attached to a given candidate regressor is set to 0, and only if the information contained in the data is valuable enough to influence the likelihood the posterior mean will be far from 0. More precisely, rather than acting as a selection device which either includes or excludes a regressor, the BVAR includes all the regressors but it assigns a different weight to each of them. The weight is higher the higher is the informational content of a given regressor. The advantage of the BVAR over the simple step-wise procedure outlined above is that the former is a fully blown approach grounded on statistical theory, while the latter is not, and as a result it might lead to incorrect inference. For example, in the simple step-wise procedure the order with which the various candidate regressors are examined can significantly influence the final outcome, and the overall size of the step-wise procedure is unknown.

BVARs have a long story in econometrics. Although the good forecasting performance of BVARs has been documented years ago by Litterman (1986) and Doan et al. (1984), only recently they have started to be used more systematically for policy analysis and forecasting macroeconomic variables (Kadiyala and Karlsson, 1997; Banbura et al., 2010; Carriero et al., 2009, 2011a,b). One of the major stumbling blocks that prevented the use of BVARs as a model for forecasting and policy analysis has typically been the large computational burden they pose. Indeed, the computation of nonlinear functions of the parameters such as impulse-response functions and multi-step forecasts need to be performed via time consuming simulations. As we will discuss below in detail, in this paper we solve this problem by directly estimating the relevant coefficients for each forecast horizon, which allows us to compute the forecasts at all forecast horizons without resorting to simulation. As a result, the production of a full set of (Bayesian) forecasts for horizon 1- to 12-month ahead takes seconds.

We compare our proposed approach against all the major forecasting models used so far in the literature, including forward rate regression (Fama and Bliss (1987), Cochrane and Piazzesi (2005)), Affine term structure Models (Ang and Piazzesi, 2003), factor models (Stock and Watson, 2002a,b), and models based on the exponential components framework (Diebold and Li, 2006; Christensen et al., 2011). For reference, we include in the comparison also a set of linear models (random walk, autoregressive models, vector autoregressions). De Pooter et al. (2007) also propose to use Bayesian techniques to forecast the term structure, but their paper differs from ours as they use Bayesian model averaging over some term structure models, while we use a large BVAR to extract efficiently the information contained in a large cross section of yields.

Besides introducing the new approach to forecasting the yields, we extend the available empirical evidence in three directions: First, for all the models considered, we provide results using homogeneous datasets, while the existing results in the literature are based on various sample periods and selected maturities. We consider two alternative datasets of US yield curves: the Fama and Bliss (1987) unsmoothed yields, publicly available on the website of the Journal of Applied Econometrics, and the smoothed yields dataset by Gurkaynak et al. (2007), publicly available on the website <http://www.federalreserve.gov>. Secondly, while most of the existing evidence evaluates forecast accuracy only in terms of statistical measures such as Root Mean Squared Forecast Errors, we also evaluate forecasts on the basis of “economic” criteria. In particular, we provide Sharpe Ratios arising from simple trading rules based on the alternative forecasts, and we use the alternative forecasts to perform optimal portfolio allocation. Finally, we provide a simulation study in which we simulate and forecast a set of “artificial” term structures. In doing such a Monte Carlo simulation the researcher has to choose the Data Generating Process (DGP). Obviously the choice of a particular DGP over another would influence the results, advantaging one rather than the other model. Therefore, rather than concentrate on simulated data and an inevitably arbitrary data generation process, we carry out our simulation by bootstrapping the actual term structure dataset. The use of a real dataset as a basis for such a robustness analysis is referred to as a ‘data based Monte Carlo method’ and discussed further in, e.g. Ho and Sørensen (1996).

We find that: (i) our proposed BVAR approach produces forecasts systematically more accurate than the random walk forecasts; (ii) the gains with respect to the random walk are small; (iii) some models beat the BVAR for a few selected maturities and forecast horizons, but they perform much worse than the BVAR in the remaining cases; (iv) predictive gains with respect to the random walk have decreased over time; (v) different loss functions (i.e., “statistical” vs. “economic”) lead to different ranking of specific models; (vi) modeling time variation in term premia is important and useful for forecasting.

The paper is structured as follows. Section 2 develops our BVAR approach. Section 3 introduces the competing forecasting models under comparison. Section 4 describes the data, the forecasting exercise and the alternative criteria we shall use in evaluating the alternative forecasts. Section 5 presents the main results, and Section 6 the robustness analysis. Finally, Section 7 concludes. To make the paper self-contained, Appendix A presents a set of technical derivations.

2. Bayesian VARs (BVAR)

The yield curve can be thought of as a vector process composed of yields of different maturities. In that light a straightforward approach to forecasting is to simply run a vector autoregression. However, such a strategy encounters an overparameterization problem, as the number of estimated parameters rapidly reduces the degrees of freedom of the VAR system. As a result, the forecasts

produced by a VAR are typically poor. To overcome this difficulty, one can either choose to sacrifice completely the cross-sectional information, and estimate, e.g. a simple AR model on each yield, or try to summarize the information in an efficient but manageable manner. This latter possibility can be pursued by modeling the system of yields as a multivariate model in which yields are a priori following a univariate autoregressive process.

2.1. Notation and preliminaries

We denote the log-price of a τ -year discount bond at time t as $p_t^{(\tau)}$, so that the log yield is $y_t^{(\tau)} = -\frac{1}{\tau}p_t^{(\tau)}$. The log forward (per-period) rate at time t for loans between $t+h$ and $t+h+\tau$ is given by $f_t(h, \tau) = \frac{1}{\tau}(-p_t^{(\tau+h)} + p_t^{(\tau)}) = \frac{1}{\tau}((\tau+h)y_t^{(\tau+h)} - \tau y_t^{(\tau)})$. The log holding period return is given by $r_{t+1}^{(\tau)} = p_{t+1}^{(\tau-1)} - p_t^{(\tau)}$. We have a cross-section of N different maturities which we denote $\tau_1, \tau_2, \dots, \tau_N$. Whenever we consider vectors of yields, returns, forward rates, or prices we use bold letters, e.g. $\mathbf{y}_t = (y_t^{(\tau_1)} y_t^{(\tau_2)} \dots y_t^{(\tau_N)})'$ is the N -dimensional vector of yields for each of the selected maturities. The h -step ahead forecasts (made at time t) are denoted with a hat, e.g. the h -step ahead forecast of the vector of yields is $\hat{\mathbf{y}}_{t+h}$.

Consider the following vector autoregression:

$$\mathbf{y}_t = \Phi_{0,h} + \Phi_{1,h}\mathbf{y}_{t-h} + \varepsilon_t; \quad \varepsilon_t \sim i.i.d.N(0, \Sigma). \quad (1)$$

Note that in the above model the vector of yields $\mathbf{y}_t$ is regressed directly onto $\mathbf{y}_{t-h}$, which means that for each forecast horizon h a different model is employed. Such an approach is known as “direct” forecasting and it focuses on minimizing the relevant loss function for each forecast horizon, i.e. the h -step ahead forecast error. Alternatively, one could use the traditional “powering up” approach, which consists in regressing $\mathbf{y}_t$ onto $\mathbf{y}_{t-1}$, and then to compute recursively the h -step ahead forecasts. For a discussion and a comparison of these alternative methods, see e.g. Marcellino et al. (2006), Pesaran et al. (2011) and Carriero et al. (2011a). In general, the powering up approach is more efficient, as it uses more datapoints, but is dangerous in the presence of misspecification, because the misspecification will inflate with the forecast horizon when the forecasts are computed recursively. On the other side, the direct approach is less efficient but is more robust to misspecification. More importantly for our purposes, the direct approach implies that the h -step ahead forecast is a linear function of the coefficients: $\hat{\mathbf{y}}_{t+h} = \hat{\Phi}_{0,h} + \hat{\Phi}_{1,h}\mathbf{y}_t$, while in the traditional powering up approach the multi-step forecasts will be a highly nonlinear function of the estimated coefficients. As a result, there is an exact closed form solution for the distribution of the h -step ahead forecasts computed using (1), while computing the forecasts resulting from the powering up strategy would require the use of time-demanding simulation methods.

2.2. A Normal-Inverted Wishart AR(1) prior

It is clear that yields, regardless of maturity, are very persistent processes. Indeed both a simple Auto Regressive (AR) model and a driftless Random Walk (RW) produce very good forecasts of the yields. Therefore it is reasonable to think a priori that each of the yields in (1) obeys a univariate AR with high persistence, or equivalently, that the expected values of the matrix $\Phi_{1,h}$ is $E[\Phi_{1,h}] = 0.99 \times I$.¹ We also need to assess how strong is the belief

¹ Alternatively, one could also estimate univariate Autoregressions for each of the yields at hand and use these estimates to elicit the prior. Such a strategy is appealing, even though a truly Bayesian approach should not use likelihood information in specifying the prior. We have also computed results using this alternative strategy. Results are similar, with the caveat that the alternative strategy produces relatively better forecasts for short maturities and relatively worse forecasts for long maturities. We do not report these results for brevity, but they are available upon request.

we have in such a prior, i.e. we need to set a variance around the prior mean. More formally, we assume that $\Phi_{1,h}$ is conditionally (on Σ) normal, with first and second moments given by:

$$E[\Phi_{1,h}^{(ij)}] = \begin{cases} 0.99 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}, \quad \text{Var}[\Phi_{1,h}^{(ij)}] = \theta \sigma_i^2 / \sigma_j^2, \quad (2)$$

where $\Phi_{1,h}^{(ij)}$ denotes the element in position (i,j) in the matrix $\Phi_{1,h}$, and where the covariances among the coefficients in $\Phi_{1,h}$ are zero. The shrinkage parameter θ measures the tightness of the prior: when $\theta \rightarrow 0$ the prior is imposed exactly and the data do not influence the estimates, while as $\theta \rightarrow \infty$ the prior becomes loose and the prior information does not influence the estimates, which will approach the standard OLS estimates. We will discuss in detail the choice of this parameter below, since it is a key element of our proposal. The factor σ_i^2 / σ_j^2 is a scaling parameter which accounts for the different scale and variability of the data. To set the scale parameters σ_i^2 we follow common practice (see e.g. Litterman, 1986; Sims and Zha, 1998) and set it equal to the variance of the residuals from a univariate autoregressive model for the variables. The prior specification is completed by assuming a Normal prior for the intercepts $\Phi_{0,h} \sim N(0, 0.001)$ and an Inverted Wishart prior for the matrix of disturbances $\Sigma \sim IW(\nu_0, S_0)$, where ν_0 and S_0 are the prior scale and shape parameters, and are set such that the prior expectation of Σ is equal to a fixed diagonal residual variance $E[\Sigma] = \text{diag}(\sigma_1^2, \dots, \sigma_N^2)$. This provides us with a Normal-Inverted Wishart ($N - IW$) prior, which is conjugate, i.e. it features a Normal-Inverted Wishart ($N - IW$) posterior distribution.

We can re-write more compactly the VAR as:

$$Y = X_h \Psi_h + E, \quad (3)$$

where $Y = [\mathbf{y}_{h+1}, \dots, \mathbf{y}_T]'$ is a $T \times N$ matrix containing all the data points in $\mathbf{y}_t$, $X_h = [1Y_{-h}]$ is a $T \times M$ matrix containing a vector of ones (1) in the first columns and the h -th lag of Y in the remaining columns, $\Psi_h = [\Phi_{0,h} \Phi_{1,h}]'$ is a $M \times N$ matrix, and $E = [\varepsilon_{h+1}, \dots, \varepsilon_T]'$ is a $T \times N$ matrix of disturbances. As only one lag is considered we have $M = N + 1$. The prior distribution can then be written as:

$$\Psi_h | \Sigma \sim N(\Psi_0, \Sigma \otimes \Omega_0), \quad \Sigma \sim IW(\nu_0, S_0). \quad (4)$$

Note that $\Psi_h | \Sigma$ is a matrixvariate normal distribution² where the prior expectation $E[\Psi_h] = \Psi_0$ and prior variance $\text{Var}[\Psi_h] = \Sigma \otimes \Omega_0$ are set according to Eq. (2). The prior above can also be conveniently implemented using dummy observations, see Banbura et al. (2010) for details.

As the used $N - IW$ prior is conjugate, the conditional posterior distribution of this model is also $N - IW$ (Tiao and Zellner, 1964; Zellner, 1973):

$$\Psi_h | \Sigma, Y \sim N(\bar{\Psi}, \Sigma \otimes \bar{\Omega}), \quad \Sigma | Y \sim IW(\bar{\nu}, \bar{S}), \quad (5)$$

where the bar denotes that the parameters are those of the posterior distribution. Defining $\hat{\Psi}$ as the OLS estimator, we have that $\bar{\Psi} = \bar{\Omega}(\Omega_0^{-1}\Psi_0 + X'Y)$, $\bar{\Omega} = (\Omega_0^{-1} + X'X)^{-1}$, $\bar{\nu} = \nu_0 + T$, and $\bar{S} = S_0 + Y'Y + \Psi_0'\Omega_0^{-1}\Psi_0 - \bar{\Psi}'\bar{\Omega}^{-1}\bar{\Psi}$. A derivation is provided in Appendix A.

2.3. Estimation

In order to perform inference and forecasting one needs the full joint posterior distribution and the marginal distributions of the parameters $\bar{\Psi}$ and Σ . One could use the conditional posteriors in (5) as a basis of a Monte Carlo algorithm that, drawing in turn from

² The matrixvariate normal distribution arises when one considers matrices of random variables. If a random matrix Z has a matrixvariate normal distribution, then its vectorisation $\text{vec}(Z)$ will have a multivariate normal distribution. A distinguishing feature of this distribution is that the variance matrix has a Kronecker structure, which implies that the variances of each random vector contained in the matrix Z are proportional one another.

the conditionals $\Sigma|Y$ and $\Psi_h|\Sigma, Y$ eventually produces a sequence of draws from the joint posterior $\Psi_h, \Sigma|Y$ and the marginal posteriors $\Psi_h|Y, \Sigma|Y$, as well as the posterior distribution of any function of these coefficients (e.g. multi-step forecasts or impulse responses).

Still, if one is interested only in the posterior distribution of Ψ_h or a linear function of it (rather than in any nonlinear function of it) there is an alternative to simulation: by integrating out Σ from (5) Zellner (1973) has shown that the marginal posterior distribution of Ψ_h is a matrix-variate t^3 :

$$\Psi_h|Y \sim MT(\bar{\Psi}, \bar{\Omega}^{-1}, \bar{S}, \bar{\nu}). \quad (6)$$

A derivation of (6) is provided in Appendix A. The expected value of this distribution is the same as the expected value of the conditional distribution $\Psi_h|\Sigma, Y$:

$$\bar{\Psi} = (\Omega_0^{-1} + X'X)^{-1} (\Omega_0^{-1}\Psi_0 + X'Y). \quad (7)$$

The matrix of coefficients $\bar{\Psi}$ is obviously extremely fast to compute, and has an intuitive interpretation. Recalling that $\hat{\Psi}$ is the OLS estimator, and using the normal equations $X'X\hat{\Psi} = X'Y$ we can rewrite it as:

$$\bar{\Psi} = (\Omega_0^{-1} + X'X)^{-1} (\Omega_0^{-1}\Psi_0 + X'X\hat{\Psi}), \quad (8)$$

which shows that the posterior mean $\bar{\Psi}$ is a weighted average of the OLS estimator and of the prior mean Ψ_0 , with weights proportional to the inverse of their respective variances. In the presence of a tight prior (i.e., when $\theta \rightarrow 0$) the posterior mean will collapse to $\bar{\Psi} = \Psi_0$, while with a diffuse prior (i.e., when $\theta \rightarrow \infty$) the posterior mean will collapse to the unrestricted OLS estimate.

Given the posterior mean $\bar{\Psi} = [\bar{\Psi}_{0,h} \ \bar{\Psi}_{1,h}]'$, it is straightforward to produce forecasts up to h steps ahead simply by setting:

$$\hat{\mathbf{y}}_{t+h} = \bar{\Phi}_{0,h} + \bar{\Phi}_{1,h}\mathbf{y}_t, \quad (9)$$

where $\hat{\mathbf{y}}_{t+h}$ is the mean of the posterior distribution of $\mathbf{y}_{t+h}$.

2.4. Prior tightness

To make the prior operational, one needs to choose the value of the hyperparameter θ . We propose to choose the value of θ by setting it to the value that maximizes the marginal data density (marginal likelihood), which can be obtained by integrating out all the model coefficients. Defining Θ as the set of all the coefficients in the model, the marginal data density is given by:

$$p_\theta(Y) = p(Y|\theta) = \int p(Y|\Theta, \theta)p(\Theta|\theta)d\Theta. \quad (10)$$

Under our Normal-Inverted Wishart prior the density $p_\theta(Y)$ can be computed in closed form (see Bauwens et al., 1999) and is given by:

$$p_\theta(Y) = \pi^{\frac{TN}{2}} \times |I + X\Omega_0(\theta)X'|^{-\frac{N}{2}} \times |S_0|^{-\frac{v_0}{2}} \times \prod_{i=1}^N \frac{\Gamma\left(\frac{v_0+T+1-i}{2}\right)}{\Gamma\left(\frac{v_0+1-i}{2}\right)} \times |S_0 + (Y - X\Psi_0)'(I + X\Omega_0(\theta)X')^{-1}(Y - X\Psi_0)|^{-\frac{v_0+T}{2}}. \quad (11)$$

At each point in time we choose θ by maximizing:

$$\theta_t^* = \arg \max_{\theta} \ln p_\theta(Y) \quad (12)$$

Note that in the expression above the prior variance $\Omega_0(\theta)$ is a function of the tightness θ . The marginal likelihood in (10) can be factorized (see e.g. Geweke, 2005, page 67) as the product of the one-step-ahead out of sample prediction record of the model.

Therefore maximizing the ML amounts to choose the tightness θ_t^* which optimizes (ex ante) the out of sample forecasting performance of the model at each forecast horizon.⁴

Also note that, when the researcher has a flat prior on θ , the data density $p_\theta(Y)$ is proportional to the posterior probability $p(\theta|Y)$:

$$p(\theta|Y) \propto p(Y|\theta)p(\theta) \propto p(Y|\theta) = p_\theta(Y). \quad (13)$$

It follows that the value of θ maximizing the marginal likelihood in (10) will also maximize the posterior in (13), if we have no a priori information on θ . Therefore the strategy proposed here can be interpreted as choosing the value of θ having the highest posterior probability. In a recent paper Giannone et al. (2012) generalize this approach to the case of a proper prior for θ , which requires an explicit hierarchical modeling and posterior simulation.

Fig. 2 depicts the value of $\ln p_\theta(Y)$ for the case with $h = 1$ as a function of θ and time. The results are based on the dataset of Fama and Bliss (1987) which is displayed in Fig. 1, and which we will describe in detail in Section 4. We consider the values of θ_t in the grid $\theta \in \{2e-16, 4e-16, 6e-16, 8e-16, 1e-15, 0.00001, 0.0001, 0.001, 0.01, 0.1, 10\}$. Interestingly, it turns out that for all forecast horizons $p_\theta(Y)$ is a hump-shaped function⁵ of θ , which shows that imposing some shrinkage (as opposed to no shrinkage or infinite shrinkage) is optimal, with the optimal value being around 0.00001. Such value looks rather low, and it implies a rather high amount of shrinkage. However one should bear in mind that the value of the shrinkage parameter θ^* is not interpretable per se but only in relation to the dimension of the system. The same value for the shrinkage parameter would imply different amounts of shrinkage if it is applied to a small or to a large system. De Mol et al. (2008) have formalized this point and have shown that the optimal shrinkage parameter θ^* goes to 0 when the cross-sectional size of the system increases.

As we estimate a different model for each forecast horizon h , we will select an optimal θ_t^* for each h . The time series path of the optimal θ_t^* for $h = 1, \dots, 12$ is depicted in Fig. 3. As is clear from the figure, the optimal θ_t^* always hovers around 0.00001 for all forecast horizons, and only for long horizons it becomes occasionally higher.⁶

3. Forecasting models

In this section we describe all the competing forecasting models under evaluation. The models include linear models (autoregressive models and vector autoregressions), forward rate regressions (along the lines of Fama and Bliss (1987) and Cochrane and Piazzesi (2005)), factor models (along the lines of Diebold and Li (2006), Christensen et al. (2011) and Stock and Watson (2002a,b)), and affine term structure models (along the lines of Ang and Piazzesi (2003)). Our proposed model, i.e. the Bayesian VAR, has been described in Section 2.

3.1. Random walk (RW)

We include a simple Random Walk forecast (RW). The RW forecast of the yield of maturity τ at time $t + h$ is given by:

$$\hat{y}_{t+h}^{(\tau)} = y_t^{(\tau)} \quad (14)$$

⁴ Note that as we use the direct approach, we use a different model for each forecast horizon. For this reason technically the forecasts produced by each model are always one-step-ahead forecasts. It follows that by optimizing the ML for each model (each corresponding to a different horizon) we are actually optimizing the forecasting performance at all horizons.

⁵ Fig. 2 depicts the natural logarithm $\ln p_\theta(Y)$, which is rather flat for intermediate values of θ , and it might seem to signal that the function is not sensitive to this parameter. However, this is a feature of the log-transformation, and the surface $p_\theta(Y)$ is much steeper.

⁶ We have also considered a fixed value for θ throughout the sample, set at 0.00001, and found that such a strategy produces very similar results to those obtained using the optimally selected θ_t^* .

³ As for the matrix-variate normal, the matrix-variate t distribution arises when one considers matrices of random variables. If a random matrix Z has a matrix-variate t distribution then its vectorisation $\text{vec}(Z)$ will have a multivariate t distribution.

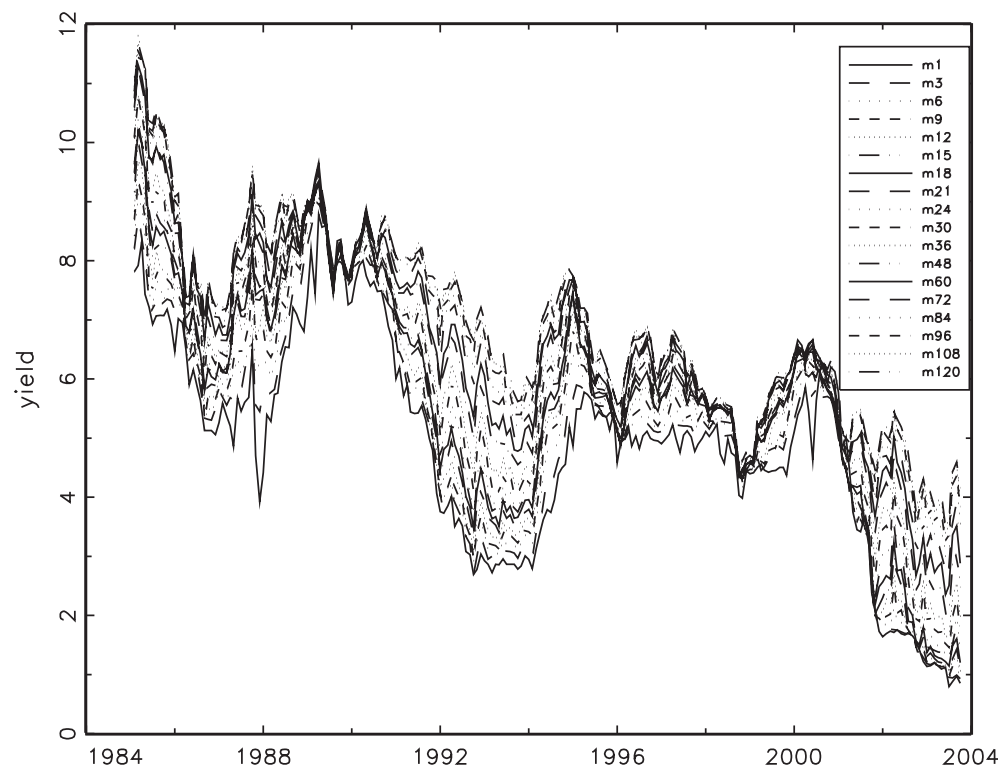


Fig. 1. US Treasury zero coupon yield curve. Data are end-of-month, unsmoothed (Fama and Bliss, 1987) zero-coupon yields of the following maturities (in months): 1, 3, 6, 9, 12, 15, 18, 21, 24, 30, 36, 48, 60, 72, 84, 96, 108, 120.

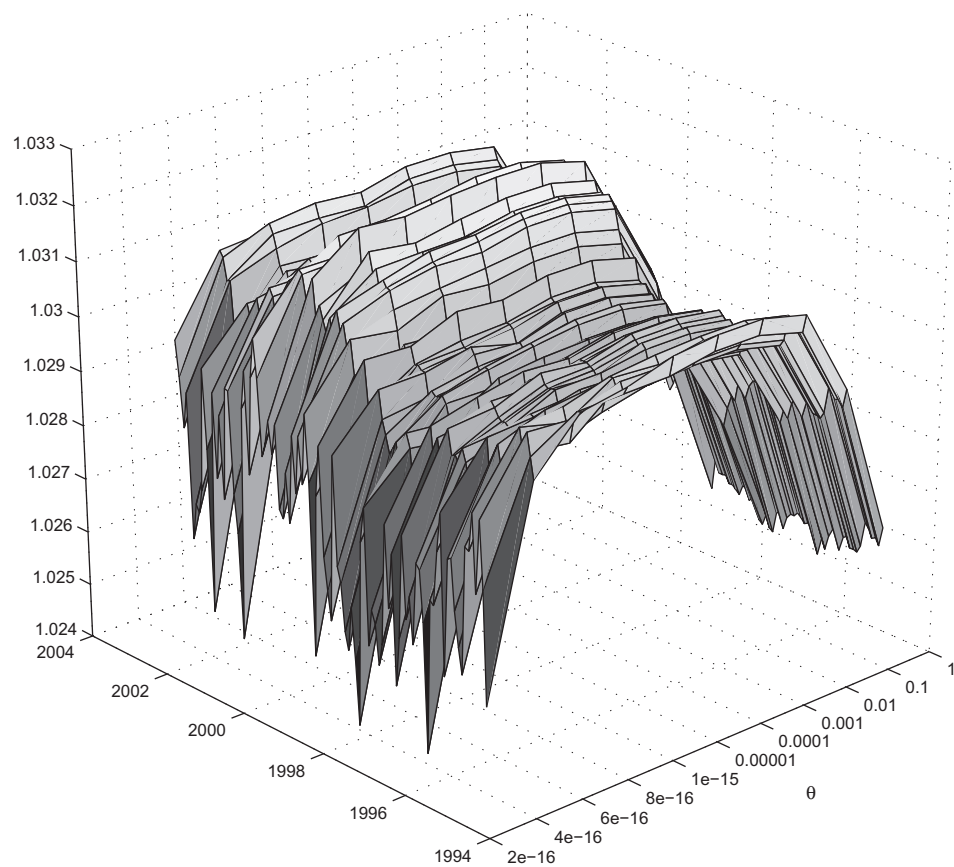


Fig. 2. Marginal Data Density. The surface represents the value of the log-data density $\ln p_{\theta}(Y) = \ln \int p(Y|\theta)p(\theta)d\theta$ for the case $h = 1$ (1-step ahead) as a function of the shrinkage parameter θ and time. To facilitate comparison across time the surface is rescaled by dividing it by the value it attains in $\theta = 10$.

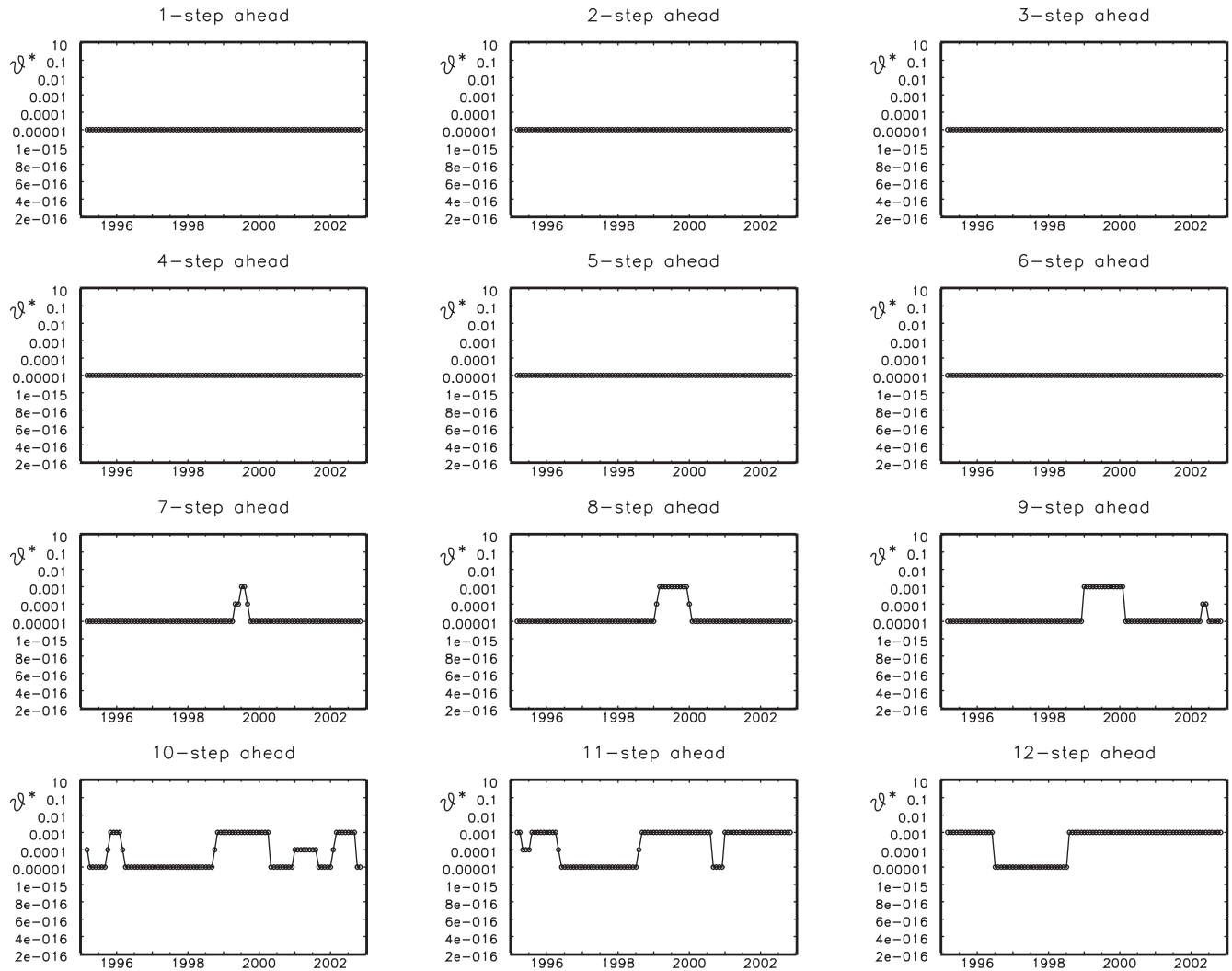


Fig. 3. Time series path of the optimal shrinkage $\theta_t^* = \arg \max_p(\theta)(Y)$ for $h = 1, \dots, 12$.

Duffee (2002) and Diebold and Li (2006) have shown that beating a Random Walk forecast of the yield curve is difficult, therefore we will use the RW forecasts as the benchmark with respect to which we will compare the forecasts of all the competing models.

3.2. Univariate Autoregressive Model (AR)

We also consider univariate autoregressive models. Forecasts from such process can be produced in two alternative ways. The first way is known as powering-up approach. In this approach the econometrician estimates the model:

$$y_t^{(\tau)} = \alpha + \beta y_{t-1}^{(\tau)} + \varepsilon_t \quad (15)$$

for the generic maturity τ . The 1-step ahead forecast is produced as $\hat{y}_{t+1}^{(\tau)} = \hat{\alpha} + \hat{\beta} y_t^{(\tau)}$, while the forecasts for h -step ahead horizon are obtained as:

$$\hat{y}_{t+h}^{(\tau)} = \hat{\alpha} + \hat{\beta} \hat{y}_{t+h-1}^{(\tau)}.$$

Alternatively, one can use the direct approach. As we discussed before, this approach directly optimizes the relevant loss function, i.e. the h -step ahead Mean Squared Forecast Error (MSFE) and has proved to be more robust but less efficient, see e.g. Marcellino et al. (2006) and Pesaran et al. (2011). The estimated model is:

$$y_t^{(\tau)} = \alpha_h + \beta_h y_{t-h}^{(\tau)} + \varepsilon_t, \quad (16)$$

i.e. the variable at time t is projected directly onto its past value in period $t - h$, so the estimated coefficients summarize the h -step

ahead effect. Note that a different α_h and β_h are obtained for each forecast horizon. The forecasts are then derived as:

$$\hat{y}_{t+h}^{(\tau)} = \hat{\alpha}_{\tau h} + \hat{\beta}_{\tau h} y_t^{(\tau)}. \quad (17)$$

We provide results for both these approaches, labeling them respectively AR (pu) and AR (di). Note that for the 1-step ahead horizon the two methods produce the same results.

Another relevant dimension of choice when using autoregressive models is the choice of the lag length. The models described above are AR(1), but in principle one could choose any lag length p . The specification with one lag is consistent with the prior mean of our BVAR, which we have described in Section 2. We have also considered richer dynamic specifications and re-estimated both the AR models (with power-up and direct approach) by choosing the lag length by means of the Bayesian Information Criterion (BIC). The chosen lag length oscillates between one and four depending on the time period, but the final results are close to those obtained with the simpler AR(1) specifications. Therefore, we present the latter results only, while we have the former available upon request.

3.3. Vector Autoregression (VAR)

VAR forecasts are produced along the same lines of univariate AR forecasts. Specifically, VAR forecasts based on powering up are produced as follows. The regression model is:

$$\mathbf{y}_t = \mathbf{A} + \mathbf{B}\mathbf{y}_{t-1} + \varepsilon_t, \quad (18)$$

where $\mathbf{y}_t = (y_t^{(\tau_1)}, y_t^{(\tau_2)}, \dots, y_t^{(\tau_N)})'$. The 1-step ahead forecast is produced as $\hat{\mathbf{y}}_{t+1} = \hat{\mathbf{A}} + \hat{\mathbf{B}}\mathbf{y}_t$, while the h -step ahead forecasts are obtained as:

$$\hat{\mathbf{y}}_{t+h} = \hat{\mathbf{A}} + \hat{\mathbf{B}}\hat{\mathbf{y}}_{t+h-1} \quad (19)$$

For the direct approach, forecasts are obtained as follows. The regression model is:

$$\mathbf{y}_t = \mathbf{A}_h + \mathbf{B}_h\mathbf{y}_{t-h} + \varepsilon_t, \quad (20)$$

where $\mathbf{y}_t = (y_t^{(\tau_1)}, y_t^{(\tau_2)}, \dots, y_t^{(\tau_N)})'$. Note that different $\mathbf{A}_h$ and $\mathbf{B}_h$ matrices are used for each forecast horizon. The forecasts are then derived as:

$$\hat{\mathbf{y}}_{t+h} = \hat{\mathbf{A}}_h + \hat{\mathbf{B}}_h\mathbf{y}_t \quad (21)$$

We provide results for both these approaches, labeling them respectively. VAR (pu) and VAR (di). Note that for the 1-step ahead horizon the two methods produce the same results. We have also experimented with longer lag specifications and BIC lag selection. This provides rather poor results which are available upon request.

3.4. Fama and Bliss (1987) (FB)

Fama and Bliss (1987) found that the spread between the τ -year forward rate and the 1-year yield predicts the 1-year excess return of the τ -year bond, with R^2 of about 18%. The natural extension of the Fama–Bliss approach in our context uses the following regression model for each maturity τ :

$$y_t^{(\tau)} - y_{t-h}^{(\tau)} = \alpha_h + \beta_h(f_{t-h}(h, \tau) - y_{t-h}^{(\tau)}) + \varepsilon_t, \quad (22)$$

where $f_t(h, \tau)$ as the forward rate at time t for loans between time $t + h$ and $t + h + \tau$. The h -step ahead forecasts $\hat{y}_{t+h}^{(\tau)}$ are then produced by projecting the change in yields from time t to time $t + h$ using the forward-spot spread at time t :

$$\hat{y}_{t+h}^{(\tau)} - y_t^{(\tau)} = \hat{\alpha}_{th} + \hat{\beta}_{th}(f_t(h, \tau) - y_t^{(\tau)}) \quad (23)$$

3.5. Cochrane and Piazzesi (2005) (CP)

Cochrane and Piazzesi (2005) show that the excess returns on one- to 5-year maturity bonds are predicted (with R^2 up to 0.44) by a single tent-shaped linear combination of the corresponding five forward rates $y_t^{(\tau=12)}, f_t(12, 12), f_t(24, 12), f_t(36, 12)$, and $f_t(48, 12)$. Although the natural extension of this approach would imply putting all the available forward rates on the right hand side of the regression, this might lead to multicollinearity when using the Gurkaynak et al. (2007) smoothed yields, as they are based on a six-parameter functional form. Therefore, we follow Cochrane and Piazzesi (2005), and use only three forward rates, $y_t^{(\tau_{12})}, f_t(24, \tau_{12})$, and $f_t(48, \tau_{12})$, in order to eliminate problems of collinearity. The forecasts are obtained as follows. For each maturity τ , we estimate the following linear regression model:

$$y_t^{(\tau)} - y_{t-h}^{(\tau)} = \alpha_h + \beta_h y_{t-h}^{(12)} + \gamma f_{t-h}(24, 12) + \delta f_{t-h}(48, 12) + \varepsilon_t \quad (24)$$

Then, forecasts are computed as:

$$\hat{y}_{t+h}^{(\tau)} - y_t^{(\tau)} = \hat{\alpha}_h + \hat{\beta}_h y_t^{(12)} + \hat{\gamma} f_t(24, 12) + \hat{\delta} f_t(48, 12) \quad (25)$$

3.6. Dynamic Nelson and Siegel model (Diebold and Li, 2006, DL)

Diebold and Li (2006) proposed a dynamic version of the Nelson and Siegel (1987) exponential components framework. They obtained very good results in terms of out of sample yield curve

forecasting. The starting point is the following yield curve interpolant:

$$y_t^{(\tau)} = \beta_1 + \beta_2 \frac{1 - e^{-\gamma\tau}}{\gamma\tau} + \beta_3 \left(\frac{1 - e^{-\gamma\tau}}{\gamma\tau} - e^{-\gamma\tau} \right), \quad (26)$$

where $\beta_1, \beta_2, \beta_3$ and γ are parameters. Diebold and Li (2006) interpreted Eq. (26) in a dynamic fashion as a latent factor model in which β_1, β_2 , and β_3 are time-varying level, slope, and curvature factors and the terms that multiply these factors are factor loadings:

$$y_t^{(\tau)} = \beta_{1t} + \beta_{2t} \frac{1 - e^{-\gamma\tau}}{\gamma\tau} + \beta_{3t} \left(\frac{1 - e^{-\gamma\tau}}{\gamma\tau} - e^{-\gamma\tau} \right). \quad (27)$$

The cross section of yields can be used to estimate via OLS⁷ the coefficients $\beta_{1t}, \beta_{2t}, \beta_{3t}$ at each point in time. Repeating the OLS regression across sections at each point in time provides a time series of estimates of the three factors $\hat{\beta}_{1t}, \hat{\beta}_{2t}, \hat{\beta}_{3t}$. Then, factors are assumed to follow a univariate autoregressive process and are forecasted as follows:

$$\hat{\beta}_{jt+h} = \hat{a}_0 + \hat{a}_1 \hat{\beta}_{jt} \quad (28)$$

where $\hat{a}_0$ and $\hat{a}_1$ are the coefficients of a regression of $\hat{\beta}_{jt}$ onto $\hat{\beta}_{jt-h}$, $j = 1, 2, 3$. Alternatively, one can assume a VAR structure for the factors, but such a strategy typically produces a worse forecasting performance given that it increases the number of parameters to be estimated without having much to gain in terms of cross-variable interaction. For completeness we have also tried a VAR(1) specification as in Diebold and Li (2006). We label respectively DL (AR) and DL (VAR) the forecasts computed using a univariate AR and a VAR specification for the factors.

Once forecasts of the factors at time $t + h$ are available, the forecast of the yields can be retrieved simply by exploiting again the cross sectional dimension of the system:

$$\hat{y}_{t+h}^{(\tau)} = \hat{\beta}_{1t+h} + \hat{\beta}_{2t+h} \frac{1 - e^{-\gamma\tau}}{\gamma\tau} + \hat{\beta}_{3t+h} \left(\frac{1 - e^{-\gamma\tau}}{\gamma\tau} - e^{-\gamma\tau} \right). \quad (29)$$

The DL model described above has proved to be empirically successful but it is based on a purely statistical approach not grounded on finance theory. In a recent paper Christensen et al. (2011) have shown how the No-Arbitrage restrictions can be imposed onto the DL model. We have also tried this specification, and we found that although it is at least as good as the standard unrestricted DL specification for small sized systems, it is generally outperformed in systems with a large cross section of yields.

3.7. Stock and Watson (2002a,b) (SW)

Stock and Watson (2002a,b) propose to overcome the curse of dimensionality problem arising in forecasting with large datasets by using a factor structure. The information contained in a set of predictors X_t is summarized by a set of K factors:

$$X_t = \Gamma F_t + u_t \quad (30)$$

where F_t is a K -dimensional multiple time series of factors and Γ a $N \times K$ matrix of loadings. The forecast for $\hat{y}_{t+h}^{(\tau)}$ given the predictors can be obtained through a two-step procedure, in which in the first step the set of predictors $\{X_t\}_{t=1}^T$ is used to estimate the factors $\{\hat{F}_t\}_{t=1}^T$ via principal components, and then the forecasts are obtained by projecting $\hat{y}_{t+h}^{(\tau)}$ onto $\hat{F}_t$ and $y_t^{(\tau)}$. In the case at hand the set of predictors used in the first step comprises the yields of all the

⁷ To do so we follow Diebold and Li (2006) and we fix the value of γ to 0.0609, which is the value that maximizes the loading on β_3 at exactly 30 months maturity. This choice enhances numerical efficiency and robustness. An alternative approach could be to estimate the model with nonlinear least squares or Maximum Likelihood.

remaining maturities. Once the factors have been derived via principal components, we estimate the following regression model:

$$y_t^{(\tau)} = \Theta_h \hat{F}_{t-h} + \beta_h y_{t-h}^{(\tau)} + \varepsilon_t, \quad (31)$$

and then the forecasts are obtained by projecting:

$$\hat{y}_{t+h}^{(\tau)} = \hat{\Theta}_h \hat{F}_t + \hat{\beta}_h y_t^{(\tau)} \quad (32)$$

Stock and Watson (2002a,b) develop theoretical results for this two-step procedure and show that under a set of moment and rank conditions the MSFE of the feasible forecast asymptotically approaches that of the optimal infeasible forecast (i.e. the one based on the knowledge of the true factors) for N and T approaching infinity, see Bai and Ng (2006) for additional details.

3.8. Affine Term Structure Models (ATSM)

We consider the model proposed by Ang and Piazzesi (2003), which is a discrete-time version of the affine class introduced by Duffie and Kan (1996), where bond prices are exponential affine functions of underlying state variables. The assumption of no arbitrage (Harrison and Kreps, 1979) guarantees the existence of a risk neutral measure Q such that the price at time t of an asset V_t that does not pay any dividends at time $t+1$ satisfies $V_t = E_t^Q[\exp(-i_t)V_{t+1}]$, where the expectation is taken under the measure Q and i_t is the short term rate. The assumption of no arbitrage is equivalent to the assumption of the existence of the Radon–Nikodym derivative ξ_{t+1} , which allows to convert the risk neutral measure to the data generating measure: $E_t^Q[\exp(-i_t)V_{t+1}] = E_t[(\xi_{t+1}/\xi_t)\exp(-i_t)V_{t+1}]$. Assume ξ_{t+1} follows a log-normal process:

$$\xi_{t+1} = \xi_t \exp(-0.5A_t' A_t - A_t' \varepsilon_{t+1}). \quad (33)$$

The A_t are called market prices of risk and are an affine function of a vector of k factors F_t :

$$A_t = A_0 + A_1 F_t, \quad (34)$$

where A_0 is a k -dimensional vector and A_1 a $k \times k$ matrix. Note that by setting $A_1 = 0$ prices of risk are constant over time. Also the short term rate is assumed to be an affine function of F_t :

$$i_t = \delta_0 + \delta_1' F_t, \quad (35)$$

where δ_0 is a scalar and δ_1 a k -dimensional vector. We assume the factors follow a zero-mean stationary vector process:

$$F_t = \Psi F_{t-1} + \Omega \varepsilon_t, \quad (36)$$

where $\varepsilon_t \sim iidN(0, \Sigma_\varepsilon)$ with $\Sigma_\varepsilon = I$ with no loss of generality. The nominal pricing kernel is defined as:

$$m_{t+1} = \exp(-i_t) \xi_{t+1} / \xi_t = \exp(-\delta_0 - \delta_1' F_t - 0.5 A_t' A_t - A_t' \varepsilon_{t+1}), \quad (37)$$

where the second equality is obtained by using (35) and (33). The nominal pricing kernel prices all assets in the economy, so by letting $p_t^{(\tau)}$ denote the time t price of a τ -period zero coupon we have:

$$p_t^{(\tau+1)} = E_t(m_{t+1} p_{t+1}^{(\tau)}). \quad (38)$$

By using the above equations is possible to show that bond prices are an affine function of the state variables:

$$p_t^{(\tau)} = \exp(\bar{A}_\tau + \bar{B}_\tau' F_t), \quad (39)$$

where $\bar{A}_\tau$ and $\bar{B}_\tau$ are a scalar and a k -dimensional vector obeying to:

$$\begin{aligned} \bar{A}_{\tau+1} &= \bar{A}_\tau + \bar{B}_\tau' (-\Omega A_0) + 0.8' \Omega \Omega' \bar{B}_\tau - \delta_0 \\ \bar{B}_{\tau+1} &= \bar{B}_\tau' (\Psi - \Omega A_1) - \delta_1', \end{aligned} \quad (40)$$

with $\bar{A}_1 = -\delta_0$ and $\bar{B}_1 = -\delta_1$. See Ang and Piazzesi (2003) for a derivation. The continuously compounded yield on a τ -period zero coupon bond is:

$$y_t^{(\tau)} = -\ln p_t^{(\tau)} / \tau = A_\tau + B_\tau' F_t, \quad (41)$$

with $A_\tau = -\bar{A}_\tau / \tau$ and $B_\tau = -\bar{B}_\tau / \tau$, so yields are also an affine function of the factors. Eqs. (36) and (41) can be interpreted as a transition and a measurement equation respectively, so the state space model for a vector of yields of N different maturities $\tau_1, \tau_2, \dots, \tau_N$ can be written as:

$$\begin{aligned} F_t &= \Psi F_{t-1} + \Omega \varepsilon_t, \\ y_t &= A + B F_t + v_t, \end{aligned} \quad (42)$$

where $A = (A_{\tau_1}, A_{\tau_2}, \dots, A_{\tau_N})'$ and $B = (B_{\tau_1}, B_{\tau_2}, \dots, B_{\tau_N})'$ are functions of the structural coefficients of the model according to Eq. (40), and where v_t is a vector of i.i.d. Gaussian measurement errors with variance Σ_v .

Following common practice we use three factors, which can be interpreted as the level, slope and curvature of the term structure. Given that scaling, shifting, or rotation of the factors provides observational equivalence, a normalization is required. Following Dai and Singleton (2000) we identify the factors by assuming factor mean equal to zero, a lower triangular structure for the matrix Ψ , and we set $\delta_1 = (1, 1, 0)$. Given this identification scheme the coefficient δ_0 equals the unconditional mean of the instantaneous rate, which can be approximated by the sample average of the 1-month yield. As for second order coefficients, we assume Ω and Σ_v to be diagonal, while we assume absence of correlation between ε_t and v_t , i.e. $E(\varepsilon_t v_t) = 0$.

We collect all the structural parameters to be estimated in the vector:

$$\vartheta = \{\Psi, \Omega, A_0, A_1, \Sigma_v\}. \quad (43)$$

In our application, we denote this model by ATSM (tvp), to emphasize that the model features time varying term premia. We also consider a specification which imposes constant risk premia, and it is obtained simply by setting $A_1 = 0$ in Eq. (34), and we label this specification ATSM (cp). We estimate ϑ with the Expectation–Maximization (EM) algorithm, evaluating the likelihood at each iteration by means of the Kalman Filter. We maximize the likelihood of the VAR with no-arbitrage restrictions using the Broyden, Fletcher, Goldfarb, and Shanno (BFGS) algorithm. ATSMs are highly nonlinear in the parameters. Maximum likelihood estimation of such models involves a difficult optimization problem, as the likelihood features several local maxima and results are very sensitive to the initial conditions. A strategy that is often adopted for estimating such models is to start from several different initial conditions, find the maximum likelihood estimate for each of them, and finally pick the highest among the detected local maxima see Duffee (2007). In our exercise we initialize our algorithm as follows. In the first estimation window we compute a maximum, then we draw 100 alternative starting points by randomizing around this maximum (drawing from a normal with variance derived from the Hessian at the maximum), maximize again, and check that none of the random initial points leads to a point with higher likelihood (i.e. a new maximum). If this is not the case, we take the new maximum and repeat the randomization until no points with higher likelihood are found. Then, for all the remaining estimation windows, we use the optimum obtained in the previous period $t-1$ as initial condition for the maximization performed in period t .

4. Forecasting the yield curve

4.1. Data

In our baseline exercise we use the end-of-month unsmoothed yields of Fama and Bliss (1987). This dataset is publicly available on the Journal of Applied Econometrics Data Archive, as part of

supplementary material to Moench (2010), and can be found at the address <http://qed.econ.queensu.ca/jae/forthcoming/moench>.

We use data going from January 1985 to September 2003. Although the available dataset starts earlier, we focus on the post-Volker period in order to avoid sample instability. We consider yields of the following 18 maturities (in months): 1, 3, 6, 9, 12, 15, 18, 21, 24, 30, 36, 48, 60, 72, 84, 96, 108, 120.

These choices provide us with a panel of 225 monthly observations on 18 different yields. Some descriptive statistics are contained in Table 1, while a plot of the data is provided in Fig. 1.

4.2. Forecasting exercise

The forecasting exercise is performed in pseudo real time, i.e. we never use information which is not available at the time the forecast is made. For computing our results we use a rolling estimation window of 120 months (10 years). We produce forecasts for all the horizons up to 12-step ahead, but for brevity we present results only for the 1-, 3-, 6-, and 12-step ahead. Results for the remaining horizons are available upon request. The initial estimation window is 1985:1–1994:12, and the initial forecast window is 1995:1–1995:12. The last estimation window is 1992:9 to 2002:9, and the last forecast window is 2002:10 to 2003:9. The choice of a rolling scheme is suggested by two reasons. First, it is a natural way to avoid problems of instability, see e.g. Pesaran and Timmermann (2005). Second, having fixed the number of observations used to compute the forecasts and therefore the resulting time series of the forecast errors allows to use the Giacomini and White (2006) test for comparing forecast accuracy. Such a test is indeed valid provided that the size of the estimation window is fixed.

4.3. Statistical loss functions

Let $\hat{y}_{t+h}^{(\tau)}(M)$ denote the forecast of $y_{t+h}^{(\tau)}$ made by model M . The Mean Squared Forecast Error (MSFE) made by model M in forecasting the yield of maturity τ at horizon h is:

$$\text{MSFE}_{\tau,h}^M = \frac{1}{T_0} \sum \left(\hat{y}_{t+h}^{(\tau)}(M) - y_{t+h}^{(\tau)} \right)^2 \quad (44)$$

where the sum is computed over all the T_0 forecasts produced. We assess predictive accuracy in terms of Relative Mean Squared Forecast Error with respect to the Random Walk:

$$\text{RMSFE}_{\tau,h}^M = \frac{\text{MSFE}_{\tau,h}^M}{\text{MSFE}_{\tau,h}^{\text{RW}}} \quad (45)$$

In order to give an idea of the absolute level of the errors, and to facilitate comparisons with other studies we shall also report the Root Mean Squared Forecast Error of the Random Walk forecast, i.e. $\sqrt{\text{MSFE}_{\tau,h}^{\text{RW}}}$. As it is customary in this literature we present results in terms of annualized percentage yields. For example, a root MSFE of 0.31 signals an error of 31 basis points in predicting the annualized yield.

To assess whether the forecasts of two competing models are statistically significantly different we use the Giacomini and White (2006) test for forecast accuracy. This is a test of equal forecasting method accuracy and as such can handle forecasts based on both nested and non-nested models, regardless from the estimation procedures used in the derivation of the forecasts, including Bayesian and semi- and non-parametric estimation methods. The test is valid so long as the number of observations used to estimate the forecasting models does not tend to infinity, which is the case in our example, as we use a rolling estimation window. We use the unconditional version of the test, which is based on the same statistic of Diebold and Mariano (1995).

4.4. Trading rules

Several papers have evaluated the forecasting performance of term structure models by looking at statistical measures such as the one discussed in the previous Section. Only a few papers (to our knowledge) have tried to use alternative loss functions to evaluate their performance. Cochrane and Piazzesi (2005) use a simple trading rule in order to evaluate the cumulative profits arising from using their proposed forward rate regression in out of sample forecasting, and Carriero and Giacomini (2011) have used a utility-based loss function in order to evaluate the degree of usefulness of the no arbitrage restrictions. In this paper we provide an assessment of the forecasting performance based on economic criteria. In particular we consider two alternative trading strategies and a portfolio utility loss function.

Note that in the forecasting exercise we use the models described in Sections 2 and 3 to produce forecasts of the yields. However, the forecasts of the yield $\hat{y}_{t+1}^{(\tau)}$ provides a forecast of the holding period return $\hat{r}_{t+1}^{(\tau+1)}$ via a simple transformation:

Table 1
Descriptive statistics, Fama–Bliss data.

Maturity	Mean	St. dev.	1st quartile	Median	3d quartile	Min	Max
1	4.88	1.84	3.74	5.02	5.89	0.79	8.78
3	5.11	1.92	3.92	5.25	6.17	0.88	9.13
6	5.24	1.94	4.02	5.36	6.32	0.96	9.32
9	5.36	1.96	4.12	5.47	6.54	0.98	9.34
12	5.51	1.99	4.38	5.60	6.84	1.04	9.68
15	5.65	2.00	4.49	5.72	7.02	1.07	9.99
18	5.74	1.99	4.49	5.79	7.10	1.15	10.19
21	5.82	1.96	4.59	5.88	7.21	1.22	10.27
24	5.86	1.93	4.58	5.90	7.26	1.30	10.41
30	6.02	1.90	4.83	6.01	7.45	1.45	10.75
36	6.14	1.85	5.02	6.12	7.55	1.62	10.79
48	6.36	1.80	5.22	6.29	7.69	2.00	11.27
60	6.47	1.75	5.31	6.38	7.70	2.35	11.31
72	6.64	1.74	5.48	6.44	7.90	2.66	11.65
84	6.73	1.68	5.58	6.57	7.95	3.00	11.84
96	6.82	1.65	5.68	6.68	8.04	3.22	11.51
108	6.87	1.65	5.68	6.72	8.08	3.39	11.66
120	6.87	1.63	5.62	6.72	8.06	3.49	11.66

Descriptive statistics. Data are end-of-month, unsmoothed (Fama and Bliss, 1987) zero-coupon yields, ranging from January 1985 to September 2003. The units are percentages on annual basis. Data include the yields of the following maturities (in months): 1, 3, 6, 9, 12, 15, 18, 21, 24, 30, 36, 48, 60, 72, 84, 96, 108, 120.

$$\hat{r}_{t+1}^{(\tau+1)} = -\tau \hat{y}_{t+1}^{(\tau)} + (\tau + 1) y_t^{(\tau+1)}. \quad (46)$$

Also note that the forecast error made in forecasting the holding period return is proportional to that made in forecasting the yield of a given bond: $\hat{r}_{t+1}^{(\tau+1)} - r_{t+1}^{(\tau+1)} = -\tau (\hat{y}_{t+1}^{(\tau)} - y_{t+1}^{(\tau)})$. This also implies that using the holding period return rather than the yields in the statistical loss function would just rescale the results.

The first trading rule is similar in spirit to that used by Carriero et al. (2009) for evaluating the performance of a portfolio of exchange rates. This rule simply suggests to buy or to short sell a bond of a given maturity depending on the sign of the forecasted variation in holding period returns. Profits at time $t + 1$ from this trading rule applied to the yield of maturity τ are given by:

$$\pi_{t,\tau}^{TR1} = \text{sign} \left(\left[\hat{r}_{t+1}^{(\tau)} - \hat{y}_{t+1}^{(1)} \right] \right) \times \left[r_{t+1}^{(\tau)} - y_{t+1}^{(1)} \right] \times 100. \quad (47)$$

As is clear, this rule uses the forecast of the excess one-period return with respect to the 1-month yield $\left[\hat{r}_{t+1}^{(\tau)} - \hat{y}_{t+1}^{(1)} \right]$ to recommend the sign of a position which is subject to the ex post return $\left[r_{t+1}^{(\tau)} - y_{t+1}^{(1)} \right] \times 100$, where 100 is a notional capital. The profits of an equally weighted portfolio composed of returns of all the maturities at hand is simply given by $\sum_{\tau} w_{\tau} \pi_{t,\tau}^{TR1}$, where w_{τ} are equal weights. Cumulative profits from time t_0 to time t can be easily computed as $\Pi_t^{TR1} = \sum_{j=t_0}^t \sum_{\tau} w_{\tau} \pi_{j,\tau}^{TR1}$.

The second trading rule is the same as in the Cochrane and Piazzesi (2005, online Appendix). Profits at time $t + 1$ from this trading rule applied to the yield of maturity τ are given by:

$$\pi_{t,\tau}^{TR2} = \left(\left[\hat{r}_{t+1}^{(\tau)} - \hat{y}_{t+1}^{(1)} \right] \times 100 \right) \times \left[r_{t+1}^{(\tau)} - y_{t+1}^{(1)} \right] \times 100. \quad (48)$$

As is clear, this rule uses the forecast of the excess one-period return with respect to the 1-month yield $\left[\hat{r}_{t+1}^{(\tau)} - \hat{y}_{t+1}^{(1)} \right]$ to recommend the size of a position which is subject to the ex post return $\left[r_{t+1}^{(\tau)} - y_{t+1}^{(1)} \right]$, where 100 is a notional capital. The recommended size of the investment is given by $\left[\hat{r}_{t+1}^{(\tau)} - \hat{y}_{t+1}^{(1)} \right] \times 100$, where 100 is the notional capital. The profits of an equally weighted portfolio composed of returns of all the maturities at hand is simply given by $\sum_{\tau} w_{\tau} \pi_{t,\tau}^{TR2}$, where w_{τ} are equal weights. Cumulative profits from time t_0 to time t can be easily computed as $\Pi_t^{TR2} = \sum_{j=t_0}^t \sum_{\tau} w_{\tau} \pi_{j,\tau}^{TR2}$.

The Sharpe ratios for a given maturity, for a generic trading rule TR used from t_0 to t can be computed as:

$$\text{Sharpe}_{\tau}^M = \frac{\bar{\pi}}{\sqrt{\frac{1}{t-t_0} \sum_{j=t_0}^t (\pi_{j,\tau}^{TR} - \bar{\pi})^2}}; \quad \bar{\pi} = \frac{1}{t-t_0} \sum_{j=t_0}^t \pi_{j,\tau}^{TR} \quad (49)$$

4.5. Optimal portfolio allocation

The trading strategies considered in the previous section provide a simple and straightforward criterion for the economic relevance of the forecasts produced by the various alternative models. Still, they focus on each yield separately and do not take explicitly into account risk-return considerations. To address these latter issues, it is interesting to see how useful the different forecasting models are when used as a basis for optimal portfolio allocation. Following Carriero and Giacomini (2011) we consider a portfolio utility loss function which directly takes into account the risk return trade-off. Consider the asset allocation problem of an investor who is buying a portfolio of N assets in period t and then sells it in period $t + 1$. In our application the relevant portfolio is a portfolio of bonds of maturities $\tau_1, \tau_2, \dots, \tau_N$ bought in period t and then sold in period $t + 1$. The vector of expected returns on this investment is given by

$E_t[\mathbf{r}_{t+1}] = E_t \left[\left(r_{t+1}^{(\tau_1+1)}, r_{t+1}^{(\tau_2+1)}, \dots, r_{t+1}^{(\tau_N+1)} \right)' \right]$. We further assume that the model does not specify conditional variance dynamics, so that the conditional variance of $\mathbf{r}_{t+1}$ at time t simply equals the unconditional variance-covariance matrix of the N assets Σ , so that $\text{Var}_t[\mathbf{r}_{t+1}] = \Sigma$. We suppose that at each time t the forecaster constructs a portfolio by choosing the weights that minimize a quadratic utility loss function:

$$w^* = \arg \min_w \left\{ -w' E_t[\mathbf{r}_{t+1}] + \frac{\gamma}{2} w' \Sigma w \right\}, \quad (50)$$

where $E_t[\cdot]$ denotes the conditional mean at time t . The classical solution (Markowitz, 1952) to this problem is given by

$$w^* = a + B E_t[\mathbf{r}_{t+1}]; \quad (51)$$

$$a = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}' \Sigma^{-1} \mathbf{1}}; \quad (52)$$

$$B = \frac{1}{\gamma} \left(\Sigma^{-1} - \frac{\Sigma^{-1} \mathbf{1} \mathbf{1}' \Sigma^{-1}}{\mathbf{1}' \Sigma^{-1} \mathbf{1}} \right), \quad (53)$$

where $\mathbf{1}$ is a $N \times 1$ vector of ones and γ is a user-defined parameter related to the coefficient of relative risk aversion δ by the relationship $\frac{\gamma}{1-\gamma} = \delta$. Our empirical results are obtained by setting $\delta = 1$, so that $\gamma = 0.5$.

Each model at hand will provide a different forecast $\hat{\mathbf{r}}_{t+1} = E_t[\mathbf{r}_{t+1}]$, therefore providing a different set of optimal weights and a different ex-post utility loss at time t given by:

$$\text{Loss}_t = -w^* (\hat{\mathbf{r}}_{t+1})' \mathbf{r}_{t+1} + \frac{\gamma}{2} w^* (\hat{\mathbf{r}}_{t+1})' \Sigma w^* (\hat{\mathbf{r}}_{t+1}). \quad (54)$$

The average ex-post utility loss from t_0 to t is given by $\frac{1}{t-t_0} \sum_{j=t_0}^t \text{Loss}_j$. Note that the Giacomini and White (2006) test can be applied to any loss function which is itself a function of the forecasts, therefore we can use it to determine whether utility losses arising from two different models are statistically significantly different or not.

5. Results

5.1. Statistical loss function

Results in terms of Relative Mean Squared Forecast Errors (RMSFE, our statistical loss function) are displayed in Table 1. The table is divided in four panels, each of them corresponding to a different forecast horizon, respectively 1, 3, 6, and 12 step ahead. Each column provides results for a different maturity. The first row in each panel contains the root MSFE of the Random Walk (RW) forecasts. All the remaining rows report the RMSFE of a given model vis a vis the RW, according to Eq. (45), so a figure smaller than one signals that the model under consideration outperforms the RW. The stars on the right of the cell entries signal the level at which the Giacomini and White (2006) test rejects the null of equal forecasting method accuracy (*, **, and *** stars mean respectively rejection at 10%, 5%, and 1% level). Note that for the 1-step ahead case the results for the AR and VAR with powering-up and direct approach produce by construction the same results and therefore they are grouped together. Several conclusions can be drawn from the table.

First, the RW confirms to be a very competitive benchmark in forecasting the term structure of bond yields. With a few exceptions at long horizons, it outperforms the AR models, which in general are a competitive benchmark in forecasting macroeconomic variables. The RW forecasts are generally more accurate than those of most of the competing models, especially at long horizons.

The BVAR is the only method that is able to systematically outperform the RW at all maturities and forecast horizons. The gains are mildly increasing with the forecast horizon, being in the range of 2–5% for the 1-step ahead horizon, 3–5% for the 3-step and the 6-step ahead horizon, and 4–5% for the 12-step ahead horizon.

Table 2
Relative Mean Square Forecast Errors, Fama-Bliss dataset.

Maturity	1-month	3-month	6-month	9-month	12-month	15-month	18-month	21-month	24-month	30-month	36-month	48-month	60-month	72-month	84-month	96-month	108-month	120-month
<i>Horizon: 1-month ahead</i>																		
RW (Root MSFE)	0.31	0.21	0.22	0.24	0.24	0.26	0.27	0.27	0.28	0.28	0.29	0.29	0.28	0.28	0.27	0.27	0.26	0.26
AR (pu, di)	1.00	1.00	0.99	1.00	1.00	1.01	1.01	1.01	1.02	1.02	1.03	1.04	1.04	1.05	1.05	1.05	1.05	1.05
AR (di)	1.00	1.00	0.99	1.00	1.00	1.01	1.01	1.01	1.02	1.02	1.03	1.04	1.04	1.05	1.05*	1.05*	1.05*	1.05
VAR (pu, di)	0.68**	0.80	0.95	1.03	1.08	1.16	1.20	1.20	1.20	1.22	1.19	1.18	1.26*	1.25*	1.26*	1.27**	1.30***	1.25**
VAR (di)	0.68**	0.80	0.95	1.03	1.08	1.16	1.20	1.20	1.20	1.22	1.19	1.18	1.26*	1.25*	1.26*	1.27**	1.30***	1.25**
BVAR	0.98	0.96	0.96	0.95	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.97	0.97	0.97	0.97	0.97	0.97	0.98
DL (AR)	0.93	1.17***	1.35***	1.28***	1.25***	1.29***	1.19***	1.19***	1.24***	1.16***	1.10**	1.09**	1.21***	1.15***	1.13***	1.06	1.05	1.05
DL (VAR)	0.76	0.81	1.02	1.11	1.10	1.15	1.11	1.14**	1.23***	1.19***	1.14**	1.13***	1.26***	1.21***	1.20***	1.10**	1.15**	1.17*
SW	0.65***	0.80	0.94	1.00	1.08	1.14*	1.11*	1.13	1.19*	1.16***	1.10**	1.07	1.15***	1.16***	1.19***	1.17***	1.14***	1.14***
FB	0.71**	0.68**	0.86	0.93	0.96	1.01	0.99	0.98	0.99	0.99	0.97	0.98	0.99	1.01	1.01	1.01	0.98	0.98
CP	1.02	0.98	1.00	1.02	1.05	1.06	1.07*	1.08*	1.12***	1.10**	1.11***	1.12***	1.19***	1.16***	1.18***	1.17***	1.16***	1.18***
ATSM (cp)	2.74***	2.79***	1.73***	1.37***	1.00	1.02	1.01	1.05	1.10*	1.09	1.04	1.08	1.14	1.19	1.26	1.26	1.40*	1.55***
ATSM (tvp)	1.04	1.20**	1.09	1.09	1.04	1.10	1.03	1.05	1.09*	1.02	0.96	0.97	1.06	1.04	1.08	1.04	1.17***	1.45***
<i>Horizon: 3-month ahead</i>																		
RW (Root MSFE)	0.56	0.48	0.49	0.51	0.52	0.55	0.55	0.56	0.58	0.57	0.57	0.56	0.55	0.53	0.52	0.50	0.49	0.48
AR (pu)	1.00	0.97	0.97	0.98	0.99	1.00	1.00	1.01	1.01	1.03	1.05	1.07	1.07	1.08	1.08	1.09	1.09	1.09
AR (di)	1.02	1.03	1.04	1.06	1.07	1.09	1.10	1.11	1.11	1.14	1.15	1.16	1.16	1.15	1.14	1.13	1.13	1.13
VAR (pu)	0.54*	0.70	0.92	1.03	1.19	1.31	1.34	1.40	1.40	1.46	1.46	1.51	1.57*	1.57*	1.59**	1.64***	1.64***	1.64***
VAR (di)	0.56	0.69	0.95	1.12	1.33	1.47	1.52	1.60	1.62*	1.67*	1.67*	1.72**	1.79***	1.77***	1.79***	1.81***	1.81***	1.79***
BVAR	0.97	0.95	0.95	0.95	0.95	0.95	0.95	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95	0.95	0.95
DL (AR)	1.11	1.30***	1.43***	1.38***	1.38***	1.35**	1.29**	1.27**	1.26*	1.19*	1.15	1.15	1.21	1.18	1.15	1.10	1.08	1.10
DL (VAR)	0.71	0.86	1.16	1.31	1.40	1.42	1.44*	1.46**	1.51***	1.49***	1.46***	1.47***	1.56***	1.51**	1.48**	1.38*	1.40*	1.45*
SW	0.52***	0.88	1.12	1.23	1.36	1.39*	1.39*	1.57*	1.70*	1.41***	1.22*	1.29*	1.50***	1.50*	1.48***	1.47**	1.38**	1.40*
FB	0.64	0.60	0.82	0.78	0.95	1.00	1.01	1.00	1.02	0.99	0.99	1.01	0.99	1.03	1.00	1.03	1.01	1.02
CP	1.09	1.08	1.12	1.15	1.19*	1.19*	1.22*	1.23*	1.26*	1.25*	1.28*	1.30*	1.37*	1.33*	1.37*	1.37*	1.36*	1.38
ATSM (cp)	0.84	0.73	0.82	0.88	0.92	0.99	0.98	0.99	1.00	0.98	0.97	1.00	1.05	1.09	1.12	1.13*	1.17	1.21*
ATSM (tvp)	1.03	1.12	1.11	1.04	1.05	1.05	1.01	1.00	1.00	0.96	0.93	0.95	1.02	1.02	1.05	1.05	1.13	1.29
<i>Horizon: 6-month ahead</i>																		
RW (Root MSFE)	0.84	0.81	0.82	0.83	0.83	0.85	0.85	0.86	0.87	0.85	0.84	0.81	0.81	0.76	0.74	0.71	0.69	0.67
AR (pu)	0.96	0.92	0.92	0.93	0.95	0.96	0.98	1.00	1.01	1.04	1.06	1.10	1.08	1.11	1.11	1.12	1.12	1.14
AR (di)	1.07	1.10	1.14	1.18	1.21	1.24	1.27	1.29	1.30	1.34	1.37	1.37	1.34	1.33	1.29	1.27	1.26	1.25
VAR (pu)	0.61	0.84	1.08	1.25	1.45	1.55	1.62	1.67	1.68	1.74	1.75	1.80	1.85*	1.89*	1.86*	1.94**	1.94**	2.00**
VAR (di)	0.83	1.18	1.58	1.84	2.16*	2.25*	2.36**	2.40**	2.41**	2.49**	2.47**	2.47**	2.51***	2.56***	2.55***	2.59***	2.58***	2.65***
BVAR	0.97	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95	0.95	0.95
DL (AR)	1.23**	1.30**	1.38**	1.39*	1.40*	1.36	1.34	1.30	1.28	1.24	1.22	1.20	1.23	1.24	1.18	1.15	1.15	1.23
DL (VAR)	1.07	1.25	1.59	1.85	1.98	2.01	2.08*	2.10*	2.14*	2.16**	2.15**	2.11**	2.10**	2.08**	1.94**	1.88*	1.89*	2.01**
SW	0.83	1.24	1.61	1.81*	1.87*	2.14*	1.96**	2.24*	2.41	1.74**	1.40	1.54	2.01***	2.17***	2.04***	2.14**	1.79**	1.81**
FB	0.93	0.79	0.78	0.98	1.08	1.07	1.07	1.06	1.07	1.05	1.04	1.06	1.04	1.08	1.04	1.06	1.04	1.07
CP	1.14	1.17*	1.21	1.25	1.31*	1.31	1.35***	1.39*	1.43*	1.43	1.46	1.48*	1.56*	1.54*	1.59*	1.57*	1.59*	1.62*
ATSM (cp)	0.67	0.70	0.82	0.90	0.97	1.00	1.01	1.00	1.00	0.99	0.99	1.02	1.05	1.11	1.11	1.13	1.16	1.23
ATSM (tvp)	1.14	1.09	1.06	1.03	1.02	1.00	0.98	0.95	0.94	0.92	0.90	0.91	0.97	1.00	1.01	1.05	1.13	1.34
<i>Horizon: 12-month ahead</i>																		
RW (Root MSFE)	1.36	1.39	1.41	1.38	1.37	1.37	1.34	1.32	1.29	1.24	1.20	1.12	1.09	1.02	0.98	0.93	0.89	0.87
AR (pu)	0.86	0.82	0.82	0.84	0.85	0.87	0.90	0.93	0.96	1.03	1.09	1.17	1.16	1.21	1.20	1.23	1.26	1.27
AR (di)	1.37	1.49	1.62	1.73	1.82	1.93	2.03	2.09	2.12	2.21	2.22	2.15	1.99	1.90	1.77	1.71	1.70	1.68*
VAR (pu)	1.02	1.18	1.33	1.45	1.57	1.64	1.71	1.77	1.79	1.85	1.88	1.96	1.99	2.07	2.05	2.17*	2.22*	2.26**
VAR (di)	2.23	2.42	2.70	2.96	3.13	3.24	3.34	3.39	3.39	3.34	3.34*	3.31*	3.27***	3.29***	3.30***	3.41***	3.47***	3.48***
BVAR	0.96	0.96	0.96	0.96	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95
DL (AR)	1.14	1.16	1.21	1.27	1.31	1.32	1.33	1.35	1.37	1.38	1.39	1.41	1.42	1.45	1.40*	1.42**	1.47***	1.55***
DL (VAR)	2.12	2.23	2.48	2.82	2.97	3.05	3.21	3.35	3.45	3.56	3.61	3.56	3.39*	3.34*	3.11*	3.09*	3.11**	3.18***
SW	1.98	2.26	2.78	2.81	2.87	3.21	3.13	3.46	3.86	3.24	2.88	2.61	3.13*	3.33**	3.19*	3.12*	3.28***	3.21***
FB	1.04***	0.99	1.01	1.00	1.03	1.04	1.05	1.04	1.04	1.03	1.01	1.04	1.00	1.07	1.06	1.05	1.04	1.00
CP	1.43	1.57	1.78	1.97	2.13	2.21	2.34	2.45	2.54	2.60	2.65	2.66	2.71*	2.64*	2.67**	2.64***	2.66***	2.73***
ATSM (cp)	0.78	0.80	0.86	0.91	0.95	0.98	0.99	1.00	1.01	1.02	1.02	1.05	1.07	1.12	1.12	1.16	1.19	1.25
ATSM (tvp)	1.10	0.97	0.90	0.89	0.88	0.86	0.85	0.85	0.85	0.84	0.84	0.86	0.91	0.96	0.99	1.07	1.20	1.39

The Table contains the Relative Mean Squared Forecast Errors vis-à-vis the Random Walk obtained by using each of the competing models, for the horizons 1-, 3, 6-, and 12-step-ahead. The evaluation sample is 1995:1 to 2003:9. For the Random Walk we report the Root Mean Squared Forecast Error. The competing models considered are Linear models estimated with powering-up and direct approach (AR(pu), AR(di), VAR(pu), and VAR(di)), Bayesian VAR (BVAR), Diebold-Li model based on AR and VAR factor structure (DL(AR) and DL(VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium. The stars on the right of the cell entries signal the level at which the [Giacomini and White \(2006\)](#) test rejects the null of equal forecasting accuracy (*, **, and *** mean respectively rejection at 10%, 5%, and 1% level).

Such gains are rather small and not statistically significant, confirming the difficulty of beating the RW benchmark, but they are systematic for all maturities and forecast horizons.

Some other models beat the BVAR for a few selected maturities and forecast horizons but they perform much worse than the BVAR in the remaining cases. Among them, it is worth mentioning the DL models, the ATSMs, and the VARs.

In the class of linear models, powering up produces overall better forecasts than the direct approach, both for ARs and for VARs. This finding suggests that the extent of mis-specification of linear models is limited. As expected the VARs are often outperformed by simpler autoregressive specifications, which is likely due to overparameterization. Still it is very interesting to note that in some cases VARs do produce better forecasts than the simple ARs. This suggests that there is some valuable information in the cross section of yields, and the problem is to extract it efficiently, as this information probably gets lost due to the overparameterization of the model.

Turning to the forward rate regressions, the FB regressions forecast quite well yields of short maturities but they have a hard time in beating the RW for yields of longer maturities, in particular for longer forecast horizons. As for the internal ranking of the two, the FB regression seems to systematically outperform the CP regression in out of sample forecasting.

Considering now the models based on the Nelson and Siegel interpolant, we find that overall the DL model based on the AR factor specification works better than the one based on the VAR factor specification, with the exception of forecasts of short maturities at short horizons, but both are outperformed by the RW.⁸

As for the ATSM models, the evidence is mixed. The model with constant risk premium, ATSM (crp) produces quite poor forecasts at the 1-step ahead horizon, then it improves at longer horizons, but it can beat the RW only for yields of short maturity. In general, the inclusion of a time varying premium improves forecast accuracy, but also the performance of the ATSM (tvp) is quite varied, depending on the forecast horizon and maturity. It produces very good forecasts for some maturities at long horizons, in some instances with gains up to 16%. However, in many other instances it produces losses with respect to the RW. We shall see how these results change when a different loss function is considered.

5.2. Trading rules and portfolio utility loss

We now consider the results in terms of our economic loss function, i.e. the two alternative trading strategies, and the portfolio utility loss. Table 3 presents the Sharpe Ratios (i.e. returns divided by standard deviation) obtained by using the trading strategies specified in (47) and (48) applied when the forecasts are obtained with our alternative methods. Note that as the strategies use only 1-step ahead forecasts, The AR and VAR models with powering-up or direct approach produce by construction the same result and are grouped together.

The results for the first trading strategy are mixed, with the RW, AR, BVAR, FB, and ATSM (tvp) providing the highest Sharpe Ratios depending on the maturity. However, it is worth noting that across all maturities the BVAR works well, producing at least the third highest Sharpe Ratio. Under the second trading strategy the BVAR is systematically outperforming the other models for any maturity longer than 21-month.

Interestingly, the ATSM (tvp) produces a rather high Sharpe ratio for both trading strategies, but importantly the presence of a time varying risk premium is essential to this result, as the ATSM (crp) produces lower and sometimes even negative returns.

In order to assess possible time variation in the performance of the models, we proceed along the lines of [Cochrane and Piazzesi \(2005, online Appendix\)](#) and we plot in Fig. 4 the cumulative ex post profits on an equally weighted portfolio containing all the yields to maturity at hand. The plot shows that the BVAR and the FB forecasts produce the highest ex post profits, outperforming the RW over time. For the second trading strategy the simple VAR forecasts also produces high profits, but only towards the end of the sample.

Finally, we consider the results for the Portfolio Utility Loss for portfolios whose weights are optimally chosen. We consider a portfolio of bonds of maturity 1, 3, 12, 24, 36, 60, and 120 months. Table 4 provides the ex-post average utility loss obtained with each of the methods at hand. The model producing the lowest average (ex-post) utility loss is the BVAR, followed by the RW and the AR. The forward rate regressions FB and CP perform quite well but not as well as the BVAR. Overall, the ATSM produce high losses, but it is confirmed the importance of allowing for time variation in risk premia, as the loss implied by the ATSM more than doubles when constant risk premia are imposed on the model.

6. Robustness

In this section we check the robustness of the results by performing the evaluation over an alternative dataset, and we conduct a simulation based on bootstrapped data that allows a better assessment of forecast uncertainty.

6.1. Gurkaynak et al. (2007) yields and sub-sample stability

The [Fama and Bliss \(1987\)](#) data are appealing as they are unsmoothed yields. Unfortunately, they are available only up to 2003. In order to check the robustness of our results, as well as to produce results on a longer time series of yields, we have repeated the analysis using the yield curve estimates by [Gurkaynak et al. \(2007\)](#). This dataset is publicly available on the website of the Federal Reserve Board, at the address <http://www.federalreserve.gov/pubs/feds/2006> and is updated regularly.

The reason we use this longer dataset as a robustness check rather than in our baseline exercise is related to the way in which such dataset is constructed.⁹ [Gurkaynak et al. \(2007\)](#) obtain the yields by using the extension by [Svensson \(1994\)](#) of the functional form that was initially proposed by [Nelson and Siegel \(1987\)](#). Such specification assumes that the forward rates are governed by six parameters, which become six factors once we consider their variation over time.

For this reason, inevitably the [Gurkaynak et al. \(2007\)](#) panel of yields is spanned by the six underlying factors, and this feature makes them less appealing than the unsmoothed data of [Fama and Bliss \(1987\)](#). However, since the function transforming the factors into yields is not linear, it still makes sense to include a large cross section of yields in a linear model such as a VAR, because the additional yields will play a role in approximating the effect of nonlinearities.

We focus on monthly data going from January 1985 to December 2008. The initial estimation window is 1985:1–1994:12, and the initial forecast window is 1995:1–1995:12. The last estimation window is 1998:1 to 2007:12, and the last forecast window is 2008:1 to 2008:12. Results based on this alternative dataset are

⁸ In order to compare the results with those in [Diebold and Li \(2006\)](#) we also considered the period going from 1994:1 to 2000:12. To produce results for the year 1994 we used a shorter rolling window of 108 months rather than 120 as in our baseline application. Results for this subsample are available upon request and are in line with those obtained by [Diebold and Li \(2006\)](#). In particular, the DL model with AR factors produces very good forecasts, and it outperforms the RW at horizons longer than 1-step ahead.

⁹ We thank an anonymous referee for pointing this out.

Table 3

Sharpe ratios for alternative models and trading strategies, Fama–Bliss dataset. The highest Sharpe ratio for each forecast horizon is highlighted in bold.

Maturity	1-Month	3-Month	6-Month	9-Month	12-Month	15-Month	18-Month	21-Month	24-Month
<i>Panel A: Trading strategy 1</i>									
RW	–	0.085	0.315	0.486	0.261	0.375	0.224	0.324	0.431
AR (pu), AR (di)	–	0.063	0.394	0.575	0.284	0.334	0.265	0.403	0.387
VAR (pu), VAR (di)	–	0.263	0.284	0.238	0.151	0.152	0.071	0.129	0.200
BVAR	–	0.209	0.441	0.521	0.314	0.270	0.282	0.322	0.431
DL (AR)	–	–0.004	0.066	0.147	0.054	–0.035	0.013	–0.013	0.045
DL (VAR)	–	0.264	0.383	0.430	0.055	–0.145	–0.038	–0.031	–0.031
SW	–	0.050	0.302	0.209	0.209	0.102	0.062	0.090	0.125
FB	–	0.421	0.463	0.467	0.286	0.200	0.205	0.270	0.368
CP	–	0.228	0.370	0.342	0.173	0.261	0.159	0.272	0.283
ATSM (cp)	–	0.130	0.137	0.141	0.307	0.240	0.192	0.194	0.223
ATSM (tvp)	–	0.029	0.334	0.434	0.294	0.314	0.350	0.286	0.291
	30-Month	36-Month	48-Month	60-Month	72-Month	84-Month	96-Month	108-Month	120-Month
RW	0.385	0.201	0.233	0.355	0.266	0.199	0.107	0.086	0.090
AR (pu), AR (di)	0.305	0.269	0.193	0.242	0.207	0.058	–0.038	0.044	0.113
VAR (pu), VAR (di)	0.197	0.183	0.171	0.150	0.107	0.042	0.084	0.010	0.026
BVAR	0.367	0.301	0.313	0.439	0.262	0.189	0.178	0.081	0.269
DL (AR)	0.027	0.077	0.035	–0.001	–0.019	–0.025	–0.021	0.060	0.018
DL (VAR)	0.055	0.032	–0.040	–0.190	–0.229	–0.188	–0.077	–0.100	–0.125
SW	0.210	0.044	0.119	0.075	–0.046	–0.115	–0.044	–0.049	–0.040
FB	0.367	0.353	0.305	0.396	0.245	0.174	0.050	0.167	0.199
CP	0.297	0.078	0.019	–0.023	–0.016	–0.064	–0.030	–0.021	0.077
ATSM (cp)	0.218	0.230	0.203	0.293	0.112	0.101	–0.023	–0.117	–0.190
ATSM (tvp)	0.292	0.357	0.330	0.192	0.065	0.028	0.104	0.014	–0.222
<i>Panel B: Trading strategy 2</i>									
	1-month	3-month	6-month	9-month	12-month	15-month	18-month	21-month	24-month
RW	–	0.211	0.372	0.446	0.274	0.252	0.218	0.305	0.431
AR (pu), AR (di)	–	0.242	0.389	0.471	0.285	0.252	0.222	0.315	0.432
VAR (pu), VAR (di)	–	0.342	0.380	0.377	0.276	0.224	0.171	0.201	0.255
BVAR	–	0.285	0.456	0.542	0.331	0.282	0.282	0.351	0.457
DL (AR)	–	–0.012	0.043	0.163	0.042	–0.030	0.025	0.048	0.084
DL (VAR)	–	0.245	0.298	0.322	0.143	0.045	0.007	–0.006	–0.024
SW	–	0.115	0.394	0.386	0.166	0.129	0.082	0.089	0.144
FB	–	0.454	0.517	0.551	0.321	0.257	0.265	0.312	0.393
CP	–	0.321	0.420	0.451	0.290	0.264	0.194	0.218	0.259
ATSM (cp)	–	0.177	0.186	0.187	0.399	0.285	0.244	0.243	0.311
ATSM (tvp)	–	0.227	0.345	0.405	0.306	0.252	0.286	0.314	0.368
	30-month	36-month	48-month	60-month	72-month	84-month	96-month	108-month	120-month
RW	0.416	0.299	0.271	0.383	0.289	0.258	0.121	0.165	0.167
AR (pu), AR (di)	0.349	0.230	0.160	0.297	0.166	0.128	0.039	0.086	0.081
VAR (pu), VAR (di)	0.222	0.208	0.182	0.176	0.100	0.117	0.108	0.091	0.109
BVAR	0.434	0.371	0.337	0.415	0.308	0.259	0.170	0.199	0.272
DL (AR)	0.112	0.108	0.038	–0.088	–0.130	–0.109	–0.056	0.009	–0.026
DL (VAR)	–0.050	–0.081	–0.120	–0.243	–0.239	–0.211	–0.113	–0.160	–0.233
SW	0.120	0.078	0.111	0.084	–0.097	–0.210	–0.062	–0.030	–0.062
FB	0.399	0.349	0.290	0.377	0.232	0.198	0.078	0.193	0.203
CP	0.200	0.066	–0.017	–0.019	–0.110	–0.070	–0.032	0.010	–0.006
ATSM (cp)	0.279	0.264	0.163	0.120	–0.027	–0.113	–0.122	–0.164	–0.242
ATSM (tvp)	0.370	0.365	0.325	0.283	0.180	0.049	0.078	0.020	–0.181

The table contains the Sharpe Ratios obtained by using each of the models at hand and the trading strategies described in Section 4.4. The evaluation sample is 1995:1 to 2003:9. The competing models considered are Random Walk (RW), Linear models estimated with powering-up and direct approach (AR (pu), AR (di), VAR (pu), and VAR (di)), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium.

contained in Table 5 (for RMSFE), Table 6 (for the trading rules) and Table 7 (for the Portfolio utility loss).

As in the Fama and Bliss (1987) dataset, the BVAR is the only method that is able to systematically outperform the RW at all maturities for short forecast horizons. The gains in this dataset are smaller than those obtained with the Fama and Bliss (1987) dataset, ranging from 0% to 2% for the 1- and 12-step ahead horizons, and in the range of 2–4% for the 3-step and the 6-step ahead horizons. Such gains are not statistically significant, but they are systematic for all maturities and forecast horizons. This result suggests that beating the RW in forecasting the yields has become even more difficult in more recent years.

Several other patterns found using the Fama and Bliss (1987) dataset are confirmed. In particular, with regards to linear models,

the powering up produces overall better forecasts than the direct approach, both for ARs and for VARs. The VARs are again outperformed by simpler autoregressive specifications, with some significant exceptions pointing at the presence of informational content in the cross section of yields. With regard to forward rate regressions, both the FB and CP regression forecast quite well yields of short maturities, with the FB regression outperforming the CP regression. The findings on the models based on the Nelson and Siegel interpolant are also very similar, with the DL (AR) producing overall better forecasts than the DL (VAR), with the exception of short maturities. As for the ATSM models, the specification with constant risk premium, ATSM (crp), produces systematically poor forecasts and the inclusion of a time varying premium improves dramatically forecast accuracy at all horizons and maturities.

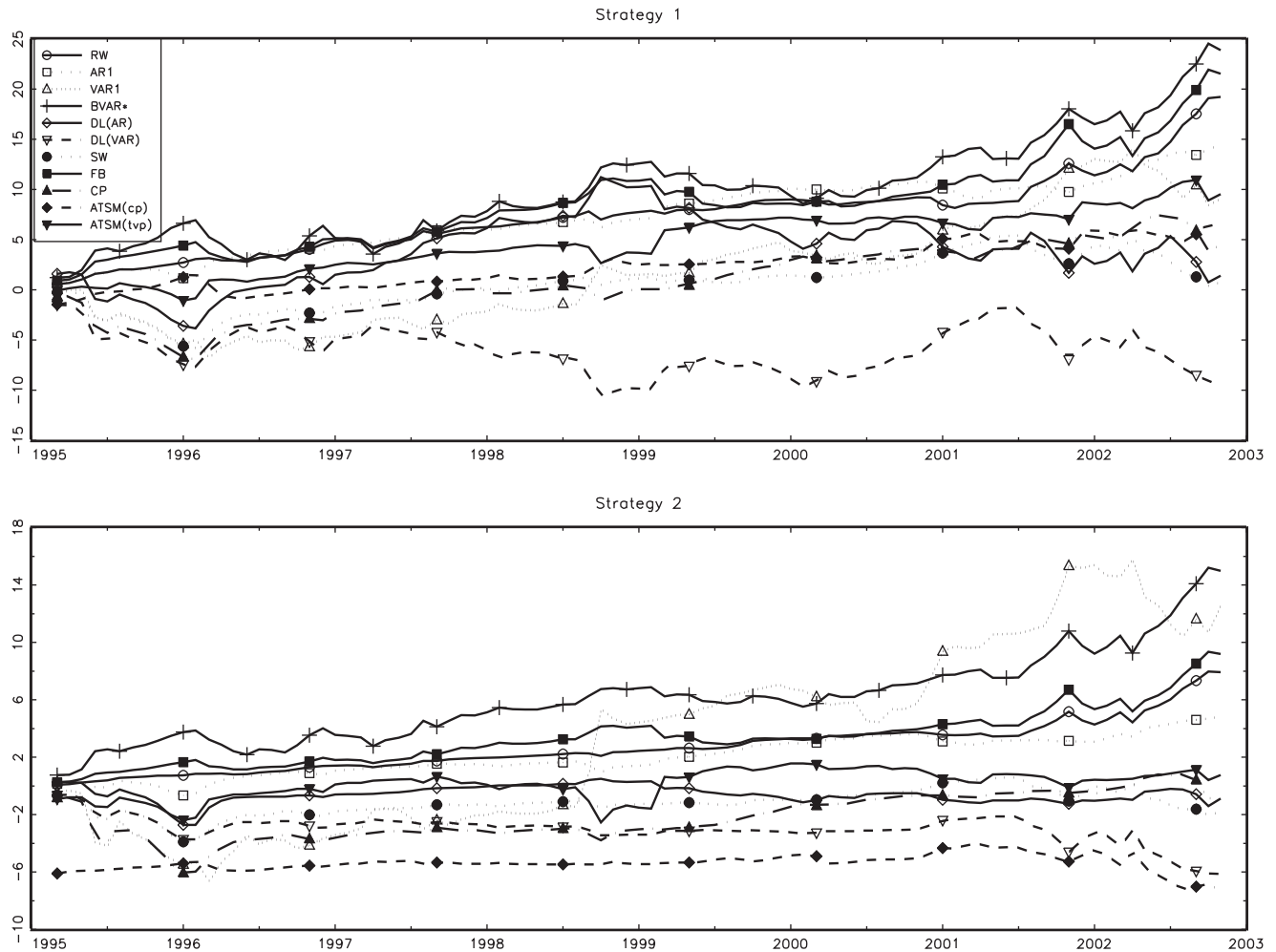


Fig. 4. Fama–Bliss dataset. Cumulative ex-post profits on an equally weighted portfolio managed using the two simple trading rules described in Section 4.4. The first panel contains results based on trading strategy 1, i.e., a trading rule that simply suggests to buy or to short sell a bond of a given maturity depending on the sign of the forecasted variation in holding period returns. The second panel contains results based on the trading strategy 2, i.e., a trading rule that uses the forecast of the excess one-period return with respect to the 1-month yield to recommend the size of a position which is subject to the ex post return. The competing models considered are Random Walk (RW), Linear models (AR and VAR), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium.

Table 4
Portfolio utility loss, Fama–Bliss data.

	Average loss	Significance
RW	−0.915	
AR	−0.821	
VAR	−0.747	
BVAR	−0.976	
dl1	0.353	***
dl2	0.341	***
SW	1.295	***
FB	−0.700	
CP	−0.608	*
ATSM (crp)	5.021	***
ATSM (tvp)	2.229	***

The table contains the utility loss obtained by using each of the models at hand and the trading strategies described in Section 4.5. The evaluation sample is 1995:1 to 2003:9. The competing models considered are Random Walk (RW), Linear models (AR, VAR), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium. The stars on the right of the cell entries signal the level at which the *Giacomini and White (2006)* test rejects the null of equal forecasting accuracy against the random walk.

*** Mean rejection at 5% level.

* Mean rejection at 10% level.

*** Mean rejection at 1% level.

The results for the trading strategies are given in Table 6.

As is clear, the BVAR provides the highest returns and Sharpe ratios in most of the cases, followed by the DL (AR) and ATSM (tvp) models. The result that the presence of a time varying risk premium is essential to improve the forecasts of the ATSM models finds additional support. Fig. 5 plots the cumulative ex post profits on an equally weighted portfolio containing all the yields to maturity at hand. The plot shows that the BVAR systematically outperforms the RW over time. Interestingly the FB regressions, which were working well in the shorter sample (see also Fig. 4), have a deterioration in the performance in the second part of the sample, while there is a marked improvement in the performance of the ATSM (tvp) model, which eventually becomes the best model under the second trading strategy. We also note that in the second part of the sample the performance of the ATSM (crp) worsens, which again points out the role of time variation in risk premia.

Finally, Table 7 provides the ex-post average utility loss obtained with each of the methods at hand. The model producing the lowest average (ex-post) utility loss is the RW, followed by the BVAR, with only a third digit difference. Note that the methods based on multivariate models perform quite poorly, because they occasionally produce highly volatile forecasts. Finally, we find additional evidence on the importance of allowing for time varia-

Table 5
Relative Mean Square Forecast Errors, Gurkaynak et al. (2007) dataset.

Maturity	1-month	3-month	6-month	9-month	12-month	15-month	18-month	21-month	24-month	30-month	36-month	48-month	60-month	72-month	84-month	96-month	108-month	120-month
Horizon: 1-month ahead																		
RW (Root MSFE)	0.31	0.21	0.20	0.22	0.23	0.24	0.26	0.26	0.27	0.28	0.29	0.29	0.29	0.28	0.27	0.27	0.26	0.25
AR (pu, di)	1.05***	1.07***	1.07***	1.07***	1.06***	1.05***	1.05***	1.04***	1.04***	1.04**	1.03*	1.03*	1.03	1.02	1.02	1.02	1.02	1.02
AR (di)	1.05***	1.07***	1.07***	1.07***	1.06***	1.05***	1.05***	1.04***	1.04***	1.04**	1.03*	1.03*	1.03	1.02	1.02	1.02	1.02	1.02
VAR (pu, di)	0.82	0.97	1.25	1.32	1.32	1.30*	1.28**	1.27**	1.25**	1.24**	1.23**	1.22***	1.23***	1.24***	1.25***	1.27***	1.29***	1.32***
VAR (di)	0.82	0.97	1.25	1.32	1.32	1.30*	1.28**	1.27**	1.25**	1.24**	1.23**	1.22***	1.23***	1.24***	1.25***	1.27***	1.29***	1.32***
BVAR	1.00	1.01	1.00	0.99	0.99	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.99
DL (AR)	0.82	1.23***	1.45***	1.38***	1.27***	1.19***	1.13***	1.10*	1.08 *	1.08	1.10	1.15**	1.17**	1.14*	1.09	1.03	1.01	1.05
DL (VAR)	0.73	0.84	1.01	1.07	1.09	1.09	1.10	1.10*	1.11 *	1.12*	1.15**	1.19***	1.22***	1.20***	1.15***	1.11***	1.10***	1.14***
SW	0.65	0.83*	1.07	1.15	1.13	1.10	1.09	1.20**	1.17***	1.14***	1.13***	1.12***	1.10*	1.08	1.10	1.15***	1.13***	1.13***
FB	0.64	0.76***	0.94	0.99	1.01	1.03	1.04	1.04	1.04*	1.05**	1.05***	1.05***	1.05***	1.05***	1.05***	1.05***	1.04***	1.04***
CP	0.92	0.84*	0.98	1.07	1.11	1.13*	1.14*	1.14*	1.14**	1.14**	1.14**	1.13**	1.13**	1.14**	1.15**	1.15***	1.16***	1.17***
ATSM (cp)	3.32***	4.02***	2.40***	1.60***	1.31***	1.21***	1.17***	1.16***	1.16***	1.16***	1.15***	1.12*	1.08	1.05	1.05	1.06***	1.10***	1.15***
ATSM (tvp)	0.77	0.89	1.08	1.17***	1.19***	1.18***	1.18***	1.17***	1.17***	1.16***	1.16**	1.14*	1.14*	1.17**	1.23***	1.32***	1.43***	1.56***
Horizon: 3-month ahead																		
RW (Root MSFE)	0.53	0.49	0.49	0.51	0.52	0.53	0.54	0.55	0.55	0.56	0.55	0.54	0.52	0.50	0.48	0.46	0.45	0.44
AR (pu)	1.13*	1.11***	1.10***	1.09**	1.08*	1.07*	1.07	1.06	1.06	1.05	1.05	1.04	1.04	1.05	1.05	1.05	1.05	1.06
AR (di)	1.12**	1.13**	1.13*	1.12*	1.12	1.11	1.11	1.10	1.10	1.09	1.09	1.09	1.09	1.09	1.09	1.09	1.09	1.09
VAR (pu)	0.63***	0.92	1.15	1.26	1.30	1.32	1.33	1.34	1.34 *	1.35*	1.36*	1.38*	1.41**	1.44***	1.47***	1.51***	1.54***	1.57***
VAR (di)	0.85	1.07	1.33	1.48	1.56*	1.60*	1.61**	1.61**	1.61***	1.62***	1.64***	1.67***	1.71***	1.74***	1.77***	1.80***	1.81***	1.81***
BVAR	0.98	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.96	0.96	0.96	0.96	0.96	0.96
DL (AR)	0.99	1.27*	1.39***	1.38**	1.33**	1.28*	1.24*	1.21	1.20	1.18	1.18	1.20	1.21*	1.19	1.15	1.09	1.05	1.02
DL (VAR)	0.63***	0.82	1.01	1.12	1.18	1.23	1.26	1.28	1.31 *	1.35**	1.40**	1.47***	1.50***	1.50***	1.46**	1.41**	1.36*	1.32 *
SW	0.68**	0.87	1.16	1.27	1.30	1.31**	1.45*	1.57*	1.46 *	1.47***	1.48***	1.49***	1.50***	1.50***	1.60***	1.53***	1.50***	1.51***
FB	0.60***	0.76	0.90	0.99	1.05	1.09	1.11	1.12	1.13	1.13*	1.13*	1.12*	1.12*	1.11*	1.10	1.10	1.09	1.09
CP	0.76*	0.90	1.11	1.24	1.33	1.39	1.42*	1.45*	1.46 *	1.48**	1.49**	1.50**	1.51**	1.51**	1.52**	1.52***	1.53***	1.53***
ATSM (cp)	1.21	1.32	1.37*	1.38*	1.38**	1.37**	1.36**	1.34**	1.33 **	1.30**	1.28*	1.23*	1.19*	1.16*	1.13*	1.12**	1.12***	1.12***
ATSM (tvp)	0.73	0.99	1.16	1.21	1.21	1.19	1.17	1.15	1.14	1.11	1.09	1.06	1.03	1.03	1.04	1.07	1.10	1.14
Horizon: 6-month ahead																		
RW (Root MSFE)	0.90	0.88	0.89	0.89	0.89	0.88	0.88	0.87	0.87	0.85	0.83	0.79	0.75	0.72	0.68	0.65	0.63	0.61
AR (pu)	1.12	1.11*	1.10	1.09	1.08	1.07	1.06	1.05	1.05	1.04	1.04	1.04	1.04	1.05	1.06	1.07	1.08	1.10
AR (di)	1.15	1.17	1.17	1.17	1.18	1.18	1.18	1.19	1.19	1.19	1.20	1.21	1.21	1.21	1.21	1.21	1.21	1.21
VAR (pu)	0.83	1.04	1.23	1.34	1.42	1.47	1.50	1.52	1.54	1.56	1.57	1.60	1.63	1.67	1.71	1.75	1.79	1.82
VAR (di)	1.29	1.74**	2.15***	2.42***	2.62***	2.76***	2.85***	2.91***	2.96***	3.01***	3.03***	3.06***	3.11***	3.17***	3.25***	3.32***	3.40***	3.47***
BVAR	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.96	0.96	0.96	0.96	0.96	0.96
DL (AR)	1.19	1.34	1.42	1.43	1.42	1.39	1.37	1.36	1.34	1.33	1.33	1.34	1.34	1.33*	1.31*	1.27	1.22	1.18
DL (VAR)	0.78	0.96	1.17	1.32	1.44	1.54	1.62	1.68	1.74 *	1.83*	1.90*	2.02*	2.08**	2.10**	2.08**	2.03*	1.98*	1.92 *
SW	0.92	0.86	1.38	1.55	1.61	1.65	2.05	2.30	2.18	2.18	2.16	2.15	2.20	2.28	2.76	2.46	2.28	2.27
FB	0.70	0.84	0.99	1.09	1.16	1.19	1.21	1.22*	1.22 *	1.21*	1.20*	1.17*	1.15*	1.13	1.12	1.10	1.10	1.09
CP	1.02	1.20	1.44	1.63	1.77	1.89	1.97	2.03	2.08 *	2.14*	2.18*	2.23*	2.26*	2.30*	2.33*	2.35*	2.37*	2.38 *
ATSM (cp)	1.23	1.36	1.44*	1.48*	1.49*	1.49*	1.48 *	1.47*	1.45*	1.42*	1.39*	1.34*	1.29*	1.26 *	1.23**	1.21***	1.20***	1.20***
ATSM (tvp)	0.89	1.02	1.10	1.12	1.10	1.08	1.05	1.02	1.00	0.96	0.93	0.91	0.90	0.92	0.95	0.99	1.04	1.08
Horizon: 12-month ahead																		
RW (Root MSFE)	1.52	1.53	1.53	1.51	1.48	1.45	1.41	1.38	1.34	1.28	1.22	1.11	1.02	0.95	0.89	0.84	0.80	0.77
AR (pu)	1.08	1.13	1.12	1.10	1.08	1.06	1.05	1.05	1.04	1.03	1.03	1.03	1.04	1.06	1.08	1.11	1.13	1.16
AR (di)	1.20	1.26	1.33	1.40	1.46	1.51	1.56	1.60	1.64	1.70	1.75	1.79	1.77	1.72	1.67	1.63	1.59	1.57
VAR (pu)	1.05	1.18	1.30	1.39	1.46	1.51	1.55	1.58	1.61	1.65	1.68	1.75	1.81	1.87	1.93	1.98	2.03	2.06 *
VAR (di)	1.91	2.21	2.47	2.64	2.77*	2.86*	2.93*	2.99*	3.03 *	3.11*	3.18*	3.32**	3.48***	3.64***	3.78***	3.91***	4.01***	4.08 ***
BVAR	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.98
DL (AR)	1.21	1.26	1.30	1.32	1.34	1.36	1.37	1.39	1.40	1.42	1.45	1.51	1.55	1.56	1.55	1.52*	1.48*	1.44 *
DL (VAR)	1.29	1.46	1.69	1.90	2.08	2.26	2.41	2.56	2.68	2.90	3.08	3.35	3.52*	3.60*	3.61*	3.57*	3.49*	3.40 *
SW	1.70	1.11	2.01	2.30	2.41	2.43	3.12*	3.71	3.82	3.80	3.73	3.70	3.82*	4.07*	5.30**	4.73	4.21*	4.17 *
FB	0.89	0.99	1.08	1.14	1.18	1.21	1.22	1.23	1.23	1.22	1.21	1.17	1.13	1.09	1.06	1.03	1.01	1.00
CP	1.82	2.09	2.40	2.66	2.88	3.06	3.20	3.32	3.42	3.56	3.67	3.83	3.95	4.03	4.09	4.13	4.14*	4.13 *
ATSM (cp)	1.47	1.48	1.49	1.50	1.51	1.52	1.53	1.53	1.53	1.53	1.52	1.50	1.47*	1.43*	1.40*	1.38**	1.36***	1.35***
ATSM (tvp)	0.86	0.88	0.89	0.88	0.86	0.84	0.82	0.80	0.78	0.77	0.76	0.79	0.86	0.94	1.04	1.14	1.24	1.33

The Table contains the Relative Mean Squared Forecast Errors vis-à-vis the Random Walk obtained by using each of the competing models, for the horizons 1-, 3, 6-, and 12-step-ahead. Results based on Gurkaynak et al. (2007) dataset. The evaluation sample is 1995:1 to 2008:12. For the Random Walk we report the Root Mean Squared Forecast Error. The competing models considered are Linear models estimated with power-up and direct approach (AR(pu), AR(di), VAR(pu), and VAR(di)), Bayesian VAR (BVAR), Diebold-Li model based on AR and VAR factor structure (DL(AR) and DL(VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium. The stars on the right of the cell entries signal the level at which the Giacomini and White (2006) test rejects the null of equal forecasting accuracy (*, **, and *** mean respectively rejection at 10%, 5%, and 1% level).

Table 6

Sharpe ratios for alternative models and trading strategies, Gurkaynak et al. (2007) dataset. The highest Sharpe ratio for each forecast horizon is highlighted in bold.

Maturity	1-Month	3-Month	6-Month	9-Month	12-Month	15-Month	18-Month	21-Month	24-Month
<i>Panel A: Trading strategy 1</i>									
RW	–	0.024	–0.007	0.058	0.108	0.131	0.128	0.125	0.120
AR (pu), AR (di)	–	0.016	0.027	–0.017	0.014	0.049	0.051	0.069	0.066
VAR (pu), VAR (di)	–	–0.015	0.079	0.122	0.115	0.064	0.070	0.094	0.060
BVAR	–	0.027	0.058	0.144	0.192	0.202	0.213	0.195	0.211
DL (AR)	–	0.122	0.013	–0.045	–0.031	0.016	0.051	0.102	0.100
DL (VAR)	–	0.149	0.045	0.031	0.001	0.065	0.044	0.101	0.116
SW	–	0.019	0.020	–0.003	–0.024	0.084	0.150	0.047	0.025
FB	–	0.118	0.171	0.196	0.174	0.130	0.125	0.123	0.072
CP	–	0.070	0.033	0.053	0.009	0.003	–0.015	–0.036	–0.048
ATSM (cp)	–	0.021	0.058	0.005	–0.073	–0.100	–0.138	–0.030	–0.011
ATSM (tvp)	–	0.189	0.105	0.000	–0.039	–0.012	0.004	0.046	0.041
	30-Month	36-Month	48-Month	60-Month	72-Month	84-Month	96-Month	108-Month	120-Month
RW0.090	0.094	0.108	0.107	0.122	0.134	0.145	0.133	0.144	
AR (pu), AR (di)	0.068	0.063	0.048	0.128	0.080	0.045	0.023	0.018	0.030
VAR (pu), VAR (di)	0.044	0.003	–0.042	–0.079	–0.106	–0.109	–0.118	–0.138	–0.119
BVAR	0.199	0.190	0.181	0.175	0.171	0.168	0.165	0.162	0.159
DL (AR)	0.077	–0.013	–0.026	–0.064	–0.090	–0.011	0.017	0.156	0.182
DL (VAR)	0.068	0.024	–0.073	–0.043	–0.077	–0.053	–0.054	–0.028	0.038
SW	–0.034	–0.078	–0.057	0.047	0.084	0.051	–0.104	–0.064	–0.096
FB	0.011	0.045	0.042	0.025	0.064	0.100	0.103	0.103	0.037
CP	–0.007	–0.032	–0.033	–0.023	–0.005	–0.022	–0.041	–0.004	0.010
ATSM (cp)	0.006	0.034	0.038	0.028	0.007	–0.063	–0.029	–0.025	0.003
ATSM (tvp)	0.063	0.078	0.051	0.034	0.133	0.122	0.120	0.134	0.151
<i>Panel B: Trading strategy 2</i>									
	1-Month	3-Month	6-Month	9-Month	12-Month	15-Month	18-Month	21-Month	24-Month
RW	–	0.090	0.094	0.122	0.149	0.166	0.175	0.179	0.179
AR (pu), AR (di)	–	0.084	0.088	0.108	0.124	0.132	0.133	0.129	0.123
VAR (pu), VAR (di)	–	0.043	–0.001	0.000	0.007	0.015	0.022	0.028	0.033
BVAR	–	0.129	0.145	0.179	0.207	0.223	0.231	0.233	0.232
DL (AR)	–	0.076	0.062	0.052	0.049	0.053	0.063	0.072	0.073
DL (VAR)	–	0.105	0.093	0.090	0.096	0.096	0.069	0.042	0.022
SW	–	0.089	0.078	0.016	–0.009	0.015	0.086	0.071	0.048
FB	–	0.212	0.171	0.171	0.164	0.151	0.135	0.119	0.104
CP	–	0.125	0.147	0.116	0.087	0.064	0.047	0.035	0.028
ATSM (cp)	–	–0.025	0.012	–0.020	–0.056	–0.071	–0.062	–0.044	–0.026
ATSM (tvp)	–	0.162	0.092	0.072	0.072	0.086	0.102	0.117	0.128
	30-Month	36-Month	48-Month	60-Month	72-Month	84-Month	96-Month	108-Month	120-Month
RW	0.175	0.169	0.157	0.148	0.142	0.136	0.132	0.129	0.126
AR (pu), AR (di)	0.109	0.097	0.079	0.068	0.062	0.058	0.055	0.049	0.041
VAR (pu), VAR (di)	0.041	0.048	0.055	0.057	0.055	0.051	0.045	0.037	0.029
BVAR	0.226	0.220	0.209	0.202	0.196	0.191	0.187	0.184	0.181
DL (AR)	0.055	0.031	0.000	–0.009	–0.005	0.014	0.063	0.139	0.133
DL (VAR)	–0.010	–0.039	–0.083	–0.099	–0.105	–0.105	–0.089	–0.036	0.007
SW	–0.066	–0.109	–0.103	–0.052	0.007	0.023	–0.127	–0.146	–0.128
FB	0.079	0.060	0.036	0.020	0.010	0.001	–0.005	–0.010	–0.014
CP	0.022	0.022	0.027	0.030	0.028	0.021	0.011	–0.001	–0.013
ATSM (cp)	–0.001	0.012	0.025	0.026	0.013	–0.051	–0.160	–0.111	–0.093
ATSM (tvp)	0.141	0.149	0.159	0.159	0.148	0.132	0.118	0.107	0.099

The table contains the Sharpe Ratios obtained by using each of the models at hand and the trading strategies described in Section 4.4. Results based on Gurkaynak et al. (2007) dataset. The evaluation sample is 1995:1 to 2008:12. The competing models considered are Random Walk (RW), Linear models estimated with powering-up and direct approach (AR (pu), AR (di), VAR (pu), and VAR (di)), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium.

tion in risk premia, as the loss implied by the ATSM doubles from 7.95 to 14.21 when constant risk premia are imposed on the model.

In order to further check for stability of our results over time, we have also run the forecasting exercise on two subsamples, each composed of seven years of forecasts (1995:1 to 2000:12 and 2001:1 to 2008:12). In order to save space, we do not report tables with results here, but they are available upon request and can be found in a previous draft of this paper. The main result emerging from the subsample analysis is that in the sample 2001:1 to 2008:12 all the models display a considerable reduction in forecast accuracy relative to the RW. The AR, which was competitive in the first subsample, becomes worse than the RW (the deterioration explains the overall performance of the AR). A similar result applies to the DL model, whose forecasting performance also deteriorates

with respect to the first subsample. Also the forecasting performance of the BVARs deteriorates, but BVARs still produce some of the best forecasts, and in several instances are still able to outperform the RW. The remaining models, i.e. the SW model and the forward rate regressions have a consistently poor performance.

6.2. An assessment with artificial data

To evaluate in more detail the performance and the robustness of the forecasting methods under analysis, we carry out a stochastic simulation in which we simulate and forecast a set of “artificial” term structures. Using several sets of artificial data allows to retrieve distributions of forecasts and forecast errors, which in turn can be used to assess the amount of variance implied by each competing model around the point forecasts. As we shall see, the sim-

ulation provides evidence that while most of the models feature an increase in forecast uncertainty with respect to the RW, the BVAR does not.

In doing a Monte Carlo simulation the researcher has to choose a Data Generating Process (DGP). Obviously the choice of a particular DGP over another would influence the results, advantaging one rather than the other model. Therefore, rather than concentrate on simulated data and an inevitably arbitrary data generation process, we carry out our simulation by bootstrapping the actual term structure dataset. The use of a real dataset as a basis for such a robustness analysis is referred to as a 'data based Monte Carlo method' and discussed further in, e.g. [Ho and Sørensen \(1996\)](#). Following this work, we create a set of artificial term structures by repeatedly bootstrapping the actual term structure.

In particular, we use the block bootstrapping algorithm described in [Politis and Romano \(1994\)](#). This algorithm is designed for block bootstrapping from a set of stationary data, therefore to implement it we first difference the data on the yields and then we recover the bootstrapped level data by adding an initial condition and integrating the artificial data. We set the initial condition equal to the first actual data point, i.e. the term structure in 1985:1. To avoid generating inappropriate patterns for the artificial term structures, we eliminate all the draws involving negative values for any of the yields at any point in time.

Table 7Portfolio utility loss, [Gurkaynak et al. \(2007\)](#) data.

	Average loss	Significance
RW	0.5714	
AR	0.8805	***
VAR	1.5134	*
BVAR	0.5782	
DL (AR)	22.2288	***
DL (VAR)	22.1051	***
SW	535.542	***
FB	0.7963	
CP	0.6402	
ATSM (crp)	14.2197	***
ATSM (tvp)	7.9544	***

The Table contains the Utility Loss obtained by using each of the models at hand and the trading strategies described in Section 4.5. Results based on [Gurkaynak et al. \(2007\)](#) dataset. The evaluation sample is 1995:1 to 2008:12. The competing models considered are Random Walk (RW), Linear models (AR, VAR), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (crp)) variation in the risk premium. The stars on the right of the cell entries signal the level at which the [Giacomini and White \(2006\)](#) test rejects the null of equal forecasting accuracy against the random walk.

** Mean rejection at 5% level.

* Mean rejection at 10% level.

*** Mean rejection at 1% level.

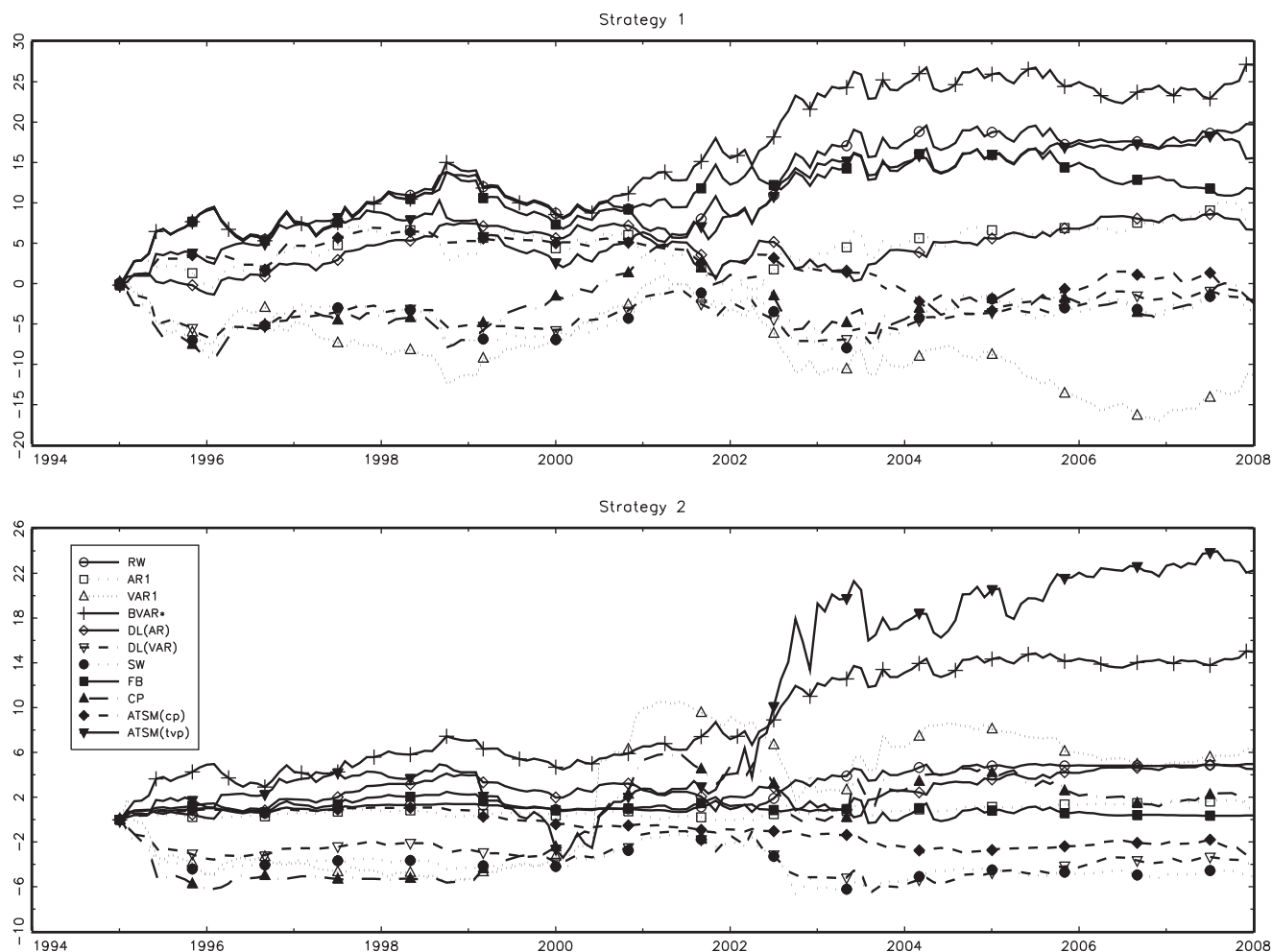


Fig. 5. [Gurkaynak et al. \(2007\)](#) dataset. Cumulative ex-post profits on an equally weighted portfolio managed using the two simple trading rules described in Section 4.4. The first panel contains results based on trading strategy 1, i.e., a trading rule that simply suggests to buy or to short sell a bond of a given maturity depending on the sign of the forecasted variation in holding period returns. The second panel contains results based on the trading strategy 2, i.e., a trading rule that uses the forecast of the excess one-period return with respect to the 1-month yield to recommend the size of a position which is subject to the ex post return. The competing models considered are Random Walk (RW), Linear models (AR and VAR), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium.

Table 8
Relative mean square forecast errors, averages over artificial samples, Fama–Bliss dataset.

Maturity	1-Month	3-Month	6-Month	9-Month	12-Month	15-Month	18-Month	21-Month	24-Month	30-Month	36-Month	48-Month	60-Month	72-Month	84-Month	96-Month	108-Month	120-month
<i>Horizon: 1-month ahead</i>																		
RW (Root MSFE)	0.36	0.28	0.29	0.30	0.33	0.33	0.33	0.34	0.34	0.34	0.35	0.35	0.35	0.34	0.33	0.32	0.32	0.33
AR (pu), AR (di)	1.02	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.03	1.02	1.03	1.02	1.03	1.03	1.03	1.03
VAR (pu), VAR (di)	1.23	1.28	1.29	1.29	1.28	1.28	1.28	1.28	1.29	1.29	1.29	1.29	1.28	1.29	1.29	1.30	1.29	1.29
BVAR	1.00	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01
DL (AR)	7.18	5.35	5.20	4.20	4.51	2.55	1.90	1.72	2.17	2.17	2.33	3.86	2.09	2.26	3.16	2.53	4.91	4.00
DL (VAR)	7.16	5.36	5.28	4.21	4.52	2.56	1.88	1.69	2.18	2.18	2.31	3.87	2.06	2.23	3.16	2.48	4.91	4.03
SW	1.30	1.08	1.08	1.09	1.08	1.07	1.10	1.09	1.09	1.09	1.10	1.08	1.09	1.10	1.09	1.09	1.09	1.09
FB	1.03	1.02	1.03	1.03	1.03	1.02	1.03	1.03	1.02	1.02	1.03	1.02	1.02	1.03	1.03	1.03	1.03	1.03
CP	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06	1.06
<i>Horizon: 3-month ahead</i>																		
RW (Root MSFE)	0.63	0.53	0.56	0.58	0.62	0.63	0.63	0.64	0.65	0.65	0.65	0.65	0.63	0.61	0.60	0.58	0.58	0.58
AR (pu)	1.06	1.06	1.06	1.06	1.06	1.05	1.05	1.05	1.05	1.05	1.05	1.05	1.05	1.05	1.06	1.06	1.06	1.06
AR (di)	1.07	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.07	1.07	1.07	1.08	1.08	1.08	1.08
VAR (pu)	1.40	1.43	1.43	1.43	1.44	1.42	1.41	1.41	1.40	1.40	1.40	1.41	1.40	1.40	1.42	1.42	1.41	1.41
VAR (di)	1.57	1.68	1.70	1.70	1.69	1.69	1.68	1.67	1.67	1.67	1.67	1.66	1.66	1.66	1.67	1.67	1.64	1.65
BVAR	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DL (AR)	3.25	2.37	2.34	2.02	2.21	1.63	1.45	1.43	1.53	1.53	1.58	2.12	1.55	1.59	1.86	1.73	2.41	2.16
DL (VAR)	3.25	2.37	2.35	2.00	2.17	1.58	1.38	1.32	1.46	1.46	1.48	2.02	1.41	1.48	1.77	1.59	2.34	2.10
SW	1.88	1.30	1.32	1.34	1.31	1.33	1.40	1.34	1.37	1.37	1.37	1.32	1.35	1.35	1.33	1.41	1.33	1.35
FB	1.07	1.08	1.08	1.10	1.09	1.07	1.08	1.09	1.08	1.08	1.09	1.07	1.07	1.08	1.07	1.08	1.08	1.09
CP	1.17	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.17	1.17	1.18	1.18
<i>Horizon: 6-month ahead</i>																		
RW (Root MSFE)	0.88	0.79	0.83	0.86	0.91	0.93	0.93	0.93	0.95	0.94	0.94	0.93	0.91	0.87	0.86	0.83	0.83	0.83
AR (pu)	1.10	1.09	1.09	1.09	1.09	1.08	1.08	1.08	1.08	1.08	1.07	1.07	1.07	1.07	1.08	1.08	1.08	1.09
AR (di)	1.14	1.16	1.15	1.15	1.15	1.15	1.14	1.15	1.14	1.14	1.14	1.13	1.13	1.13	1.15	1.14	1.14	1.15
VAR (pu)	1.53	1.54	1.54	1.54	1.53	1.51	1.50	1.49	1.48	1.48	1.48	1.48	1.48	1.47	1.49	1.49	1.48	1.49
VAR (di)	1.97	2.02	2.02	2.01	1.99	1.99	1.98	1.98	1.97	1.97	1.95	1.94	1.93	1.93	1.93	1.95	1.90	1.89
BVAR	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
DL (AR)	2.38	1.90	1.86	1.70	1.82	1.56	1.50	1.53	1.56	1.56	1.60	1.88	1.59	1.58	1.69	1.68	1.92	1.87
DL (VAR)	2.34	1.85	1.77	1.58	1.67	1.39	1.30	1.28	1.35	1.35	1.35	1.62	1.31	1.33	1.46	1.41	1.73	1.68
SW	3.02	1.80	1.79	1.82	1.78	1.79	1.88	1.79	1.86	1.86	1.84	1.75	1.77	1.84	1.76	1.92	1.74	1.81
FB	1.15	1.16	1.18	1.18	1.16	1.16	1.18	1.16	1.16	1.16	1.18	1.15	1.15	1.16	1.16	1.16	1.16	1.16
CP	1.34	1.36	1.36	1.35	1.35	1.35	1.35	1.35	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.33	1.34	1.35
<i>Horizon: 12-month ahead</i>																		
RW (Root MSFE)	1.24	1.15	1.20	1.24	1.30	1.32	1.32	1.33	1.35	1.33	1.32	1.30	1.27	1.22	1.21	1.16	1.16	1.16
AR (pu)	1.12	1.14	1.13	1.14	1.13	1.12	1.12	1.12	1.11	1.11	1.11	1.11	1.11	1.11	1.12	1.12	1.11	1.13
AR (di)	1.29	1.31	1.29	1.30	1.29	1.29	1.28	1.28	1.28	1.28	1.29	1.28	1.29	1.28	1.30	1.30	1.29	1.30
VAR (pu)	1.71	1.74	1.75	1.75	1.73	1.70	1.68	1.66	1.66	1.64	1.65	1.64	1.63	1.61	1.62	1.65	1.64	1.64
VAR (di)	2.11	2.16	2.14	2.11	2.05	2.06	2.04	2.03	2.04	2.01	2.00	1.97	1.99	2.00	1.99	2.03	2.02	2.03
BVAR	0.99	0.99	0.99	0.99	1.00	1.00	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.98	0.99
DL (AR)	2.11	1.92	1.88	1.76	1.85	1.74	1.74	1.80	1.82	1.82	1.86	2.02	1.88	1.85	1.83	1.91	1.90	1.96
DL (VAR)	1.96	1.74	1.62	1.47	1.53	1.39	1.34	1.33	1.38	1.38	1.38	1.51	1.34	1.34	1.39	1.39	1.50	1.54
SW	5.68	2.87	2.57	2.46	2.49	2.49	2.69	2.62	2.69	2.69	2.59	2.39	2.37	2.57	2.39	2.68	2.57	2.47
FB	1.36	1.32	1.33	1.37	1.39	1.32	1.34	1.33	1.32	1.32	1.35	1.29	1.31	1.33	1.32	1.32	1.37	1.29
CP	1.73	1.69	1.69	1.69	1.68	1.67	1.69	1.69	1.69	1.69	1.69	1.70	1.70	1.70	1.69	1.69	1.70	1.72

The table contains the average Relative Mean Squared Forecast Errors vis-à-vis the Random Walk over 100 bootstrap replications, obtained by using each of the competing models and the data based Monte Carlo method described in Section 6.2. For the Random Walk we report the average Root Mean Squared Forecast Error over the 100 replications. The competing models considered are Linear models estimated with powering-up and direct approach (AR (pu), AR (di), VAR (pu), and VAR (di)), Bayesian VAR (BVAR), Diebold–L i model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium. The stars on the right of the cell entries signal the level at which the [Giacomini and White \(2006\)](#) test rejects the null of equal forecasting accuracy (*, **, and *** mean respectively rejection at 10%, 5%, and 1% level).

Table 9

Sharpe ratios, averages over artificial samples, Fama–Bliss dataset. The highest Sharpe ratio for each forecast horizon is highlighted in bold.

Maturity	1-Month	3-Month	6-Month	9-Month	12-Month	15-Month	18-Month	21-Month	24-Month
<i>Panel A: Trading strategy 1</i>									
RW	–	1.52	1.44	1.30	1.38	0.91	0.74	0.72	0.84
AR	–	1.52	1.44	1.30	1.37	0.90	0.73	0.70	0.83
VAR	–	1.48	1.38	1.17	1.20	0.74	0.59	0.57	0.69
BVAR	–	1.52	1.44	1.30	1.37	0.91	0.74	0.71	0.84
DL (AR)	–	0.50	0.32	0.19	0.19	0.04	0.15	0.30	0.36
DL (VAR)	–	0.61	0.39	0.26	0.20	0.04	0.13	0.29	0.36
SW	–	1.51	1.43	1.28	1.36	0.86	0.68	0.65	0.79
FB	–	1.54	1.46	1.31	1.37	0.90	0.73	0.70	0.82
CP	–	1.51	1.43	1.29	1.36	0.88	0.71	0.68	0.81
	30-Month	36-Month	48-Month	60-Month	72-Month	84-Month	96-Month	108-Month	120-Month
RW	0.94	0.88	1.31	0.83	0.86	1.14	0.91	1.40	1.21
AR	0.92	0.87	1.30	0.81	0.85	1.12	0.88	1.38	1.20
VAR	0.78	0.76	1.12	0.68	0.71	0.96	0.76	1.21	1.08
BVAR	0.94	0.89	1.32	0.83	0.86	1.13	0.91	1.39	1.21
DL (AR)	0.43	0.26	0.41	0.21	0.19	0.12	0.09	–0.07	–0.11
DL (VAR)	0.46	0.24	0.39	0.17	0.14	0.14	0.02	–0.07	–0.03
SW	0.89	0.85	1.27	0.77	0.81	1.10	0.85	1.34	1.18
FB	0.92	0.88	1.31	0.82	0.86	1.14	0.90	1.40	1.21
CP	0.91	0.85	1.27	0.78	0.81	1.10	0.88	1.36	1.18
<i>Panel B: Trading strategy 2</i>									
	1-Month	3-Month	6-Month	9-Month	12-Month	15-Month	18-Month	21-Month	24-Month
RW	–	2.28	1.93	1.63	1.64	0.99	0.77	0.75	0.89
AR	–	2.27	1.92	1.62	1.63	0.98	0.76	0.73	0.88
VAR	–	2.25	1.88	1.54	1.55	0.89	0.66	0.63	0.78
BVAR	–	2.28	1.93	1.63	1.64	0.98	0.77	0.73	0.88
DL (AR)	–	0.74	0.47	0.30	0.28	0.04	0.15	0.34	0.37
DL (VAR)	–	0.84	0.55	0.33	0.27	0.04	0.12	0.30	0.37
SW	–	2.25	1.91	1.61	1.61	0.95	0.72	0.69	0.86
FB	–	2.29	1.93	1.62	1.62	0.98	0.75	0.73	0.88
CP	–	2.27	1.92	1.62	1.62	0.96	0.74	0.71	0.87
	30-Month	36-Month	48-Month	60-Month	72-Month	84-Month	96-Month	108-Month	120-Month
RW	1.00	0.95	1.48	0.88	0.94	1.26	0.98	1.62	1.42
AR	0.99	0.94	1.48	0.87	0.93	1.25	0.97	1.61	1.41
VAR	0.92	0.85	1.41	0.78	0.83	1.16	0.89	1.54	1.35
BVAR	0.99	0.94	1.48	0.89	0.93	1.26	0.98	1.61	1.42
DL (AR)	0.47	0.28	0.46	0.21	0.22	0.11	0.11	–0.09	–0.13
DL (VAR)	0.51	0.23	0.44	0.20	0.16	0.15	0.03	–0.04	–0.05
SW	0.97	0.91	1.46	0.85	0.89	1.24	0.96	1.60	1.40
FB	0.99	0.94	1.48	0.87	0.92	1.26	0.97	1.61	1.42
CP	0.98	0.92	1.47	0.85	0.91	1.24	0.97	1.60	1.40

The Table contains the average Sharpe Ratios over 100 bootstrap replications, obtained by using each of the models at hand and the trading strategies described in Section 4.4, and the data based Monte Carlo method described in Section 6.2. The competing models considered are Random Walk (RW), Linear models (AR, VAR), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium. The stars on the right of the cell entries signal the level at which the [Giacomini and White \(2006\)](#) test rejects the null of equal forecasting accuracy (*, **, and *** mean respectively rejection at 10%, 5%, and 1% level).

We repeat this procedure 100 times, using the [Fama and Bliss \(1987\)](#) dataset, and then for each of the 100 repetitions we implement our forecasting exercise. We do not include in the experiment the ATSM models, as these models are highly nonlinear and their estimation makes the dimension of the problem unmanageable. This provides us with 100 time series of forecasts for each yield and forecast horizon, which can be used to compute 100 RMSFEs (one for each repetition) for each model. Finally, the RMSFEs are averaged through the repetitions to get the average RMSFE made by model M in predicting h -step ahead the yield of maturity τ . The square roots of these, denoted $\text{avgRMSFE}_{\tau,h}^M$, are displayed in [Table 8](#). Similarly, we can define the average of the Sharpe Ratio from a given trading strategy based on model M . These average Sharpe ratios are displayed in [Table 9](#). Finally, we can similarly produce the average portfolio utility loss by averaging the loss in [\(54\)](#) over the 100 replications. Results for this case are in [Table 10](#).

Looking at the first row of [Table 8](#), i.e. the Average Root MSFE of the RW, it is in line with that reported for the actual data (see [Ta-](#)

[ble 2](#)). Interestingly all the models but the BVARs feature much higher average MSFE than the RW. As the bias for all the models is small,¹⁰ this implies that the variance of the forecasts of all the models at hand tends to be higher than that of the RW. The only exception is the BVAR, whose forecast variability is reduced by the shrinkage towards the AR forecasts. The BVAR produces results very close to the RW, with slight losses at one-step-ahead and very small gains at longer horizons.

Similarly, [Table 9](#) shows that the BVAR and RW are producing the best Sharpe ratios in the majority of cases. Finally, [Table 10](#) shows that in the artificial samples all the models but the DL produce similar results, with RW, BVAR and AR being slightly better.

Very similar results are obtained by using the [Gurkaynak et al. \(2007\)](#) dataset to simulate the artificial yields. in order to save

¹⁰ For brevity we do not report results on the biases, which are available upon request.

Table 10

Utility loss, artificial samples, Fama–Bliss data.

	Average loss
RW	–116.776
AR	–116.621
VAR	–115.838
BVAR	–116.767
DL (AR)	–0.075
DL (VAR)	–0.075
SW	–113.692
FB	–115.622
CP	–116.535

The Table contains the average Utility Loss over 100 bootstrap replications, obtained by using each of the models at hand and the trading strategies described in Section 4.5, and the data based Monte Carlo method described in Section 6.2. The competing models considered are Random Walk (RW), Linear models (AR, VAR), Bayesian VAR (BVAR), Diebold–Li model based on AR and VAR factor structure (DL (AR) and DL (VAR)), Stock and Watson factor model (SW), forward rate regressions (FB and CP), and ATSM with (ATSM (tvp)) and without (ATSM (cp)) variation in the risk premium. The stars on the right of the cell entries signal the level at which the Giacomini and White (2006) test rejects the null of equal forecasting accuracy against the random walk (*, **, and *** mean respectively rejection at 10%, 5%, and 1% level).

space, we do not report these results here, but they are available upon request and can be found in a previous draft of this paper.

7. Conclusions

In this paper we introduce a new statistical model for the entire term structure of interest rates and compare its forecasting performance with the current most promising alternatives, evaluating the forecasts in terms of both statistical and economic criteria.

Our proposal is a large Bayesian Vector Autoregression (BVAR), with time-varying optimal shrinkage towards a set of univariate AR models. We choose the optimal amount of shrinkage by maximizing the marginal likelihood of the model. The underlying idea is that, while in general it is difficult to outperform univariate AR or Random Walk models, there are time periods with enhanced interaction among yields at different maturities, and capturing such an interaction only when it is indeed at work further enhances the forecasting performance.

We suggest a prior for the BVAR that, combined with the formulation of direct rather than iterated forecasts, eliminates the need of time-demanding simulation methods. Therefore, one can estimate the BVAR and forecast the entire term structure for different horizons in a few seconds.

We then compare the forecasting performance of our BVAR with that of a large set of alternative models for the term structure of government bond yields. Specifically, we consider linear models (random walk, autoregressive models, vector autoregressions), factor models (Diebold and Li, 2006; Christensen et al., 2011; Stock and Watson, 2002a,b), forward rate regression (Fama and Bliss, 1987; Cochrane and Piazzesi, 2005), and Affine term structure Models (Ang and Piazzesi (2003)).

Besides introducing the new approach to forecasting the yields, we extend the available empirical evidence in three directions: (i) For all the models considered we provide results for two homogeneous datasets, while the existing results in the literature are based on various sample periods and selected maturities. We use both an updated Fama and Bliss (1987) dataset and the novel dataset of Gurkaynak et al. (2007). (ii) While most of the existing evidence evaluates forecast accuracy only in terms of statistical measures such as Root Mean Squared Forecast Errors, we also evaluate forecasts on the basis of “economic” criteria. In particular we provide Sharpe Ratios arising from simple trading rules based on the alternative forecasts, and we use the alternative forecasts to perform

optimal portfolio allocation. (iii) We provide an extensive robustness check of the results using subsample analysis and a Monte Carlo study in which we simulate and forecast a set of “artificial” term structures. Using several sets of artificial data allows to retrieve distributions of forecasts and forecast errors, which in turn can be used to assess the amount of variance implied by each competing model around the point forecasts.

Our main empirical results can be summarized as follows: (i) our proposed BVAR approach produces forecasts systematically more accurate than the random walk forecasts; (ii) the gains with respect to the random walk are small; (iii) some models beat the BVAR for a few selected maturities and forecast horizons, but they perform much worse than the BVAR in the remaining cases; (iv) predictive gains with respect to the random walk have decreased over time; (v) different loss functions (i.e., “statistical” vs. “economic”) lead to different ranking of specific models; (vi) modeling time variation in term premia is important and useful for forecasting.

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Appendix A

In order to make the paper self-contained, we report here the derivation of the posteriors and marginal likelihood of the BVAR. A more detailed analysis of this model can be found in Tiao and Zellner (1964), Zellner (1973), or Geweke (2005).

A.1. Likelihood function

Recall the VAR in Eq. (3):

$$Y = X_h \Psi_h + E, \quad (55)$$

where $Y = [y_{h+1}, \dots, y_T]'$ is a $T \times N$ matrix containing all the data points in y_t , $X_h = [1Y_{-h}]$ is a $T \times M$ matrix containing a vector of ones in the first column and the h th lag of Y in the remaining columns, $\Psi_h = [\Phi_{0,h} \Phi_{1,h}]'$ is a $M \times N$ matrix, and $E = [e_{h+1}, \dots, e_T]'$ is a $T \times N$ matrix of disturbances with variance $\Sigma_h \otimes I$. As h simply indicates the horizon for which the model is estimated in what follows we omit the subscript h in order to economize on notation.

The likelihood of this VAR is:

$$P(Y|\Psi, \Sigma) = 2\pi^{-0.5NT} |\Sigma|^{-0.5T} \exp\{-0.5\text{tr}[\Sigma^{-1}(Y - X\Psi)'(Y - X\Psi)]\} \quad (56)$$

This can be written as:

$$P(Y|\Psi, \Sigma) \propto |\Sigma|^{-0.5M} \exp\{-0.5\text{tr}[\Sigma^{-1}(\Psi - \hat{\Psi})'X'X(\Psi - \hat{\Psi})]\} \times |\Sigma|^{-0.5(T-M)} \exp\{-0.5\text{tr}[\Sigma^{-1}(Y - X\hat{\Psi})'(Y - X\hat{\Psi})]\}, \quad (57)$$

where $\hat{\Psi}$ is the OLS estimator.¹¹ Eq. (57) shows that the likelihood can be factorised as the product of the kernels of a Wishart marginal distribution for Σ^{-1} (second line) and a conditional (on Σ) matrix-

¹¹ To go from (56) and (57) one needs to rewrite the term in the brackets as follows:

$$\begin{aligned} (Y - X\Psi)'(Y - X\Psi) &= ((Y - X\hat{\Psi}) - X(\Psi - \hat{\Psi}))'((Y - X\hat{\Psi}) - X(\Psi - \hat{\Psi})) \\ &= (Y - X\hat{\Psi})'(Y - X\hat{\Psi}) - (Y - X\hat{\Psi})'X(\Psi - \hat{\Psi}) - (Y - X\hat{\Psi})'X(\Psi - \hat{\Psi}) \\ &\quad + (Y - X\hat{\Psi})'X'X(\Psi - \hat{\Psi}) \\ &= (Y - X\hat{\Psi})'(Y - X\hat{\Psi}) + (Y - X\hat{\Psi})'X'X(\Psi - \hat{\Psi}), \end{aligned}$$

where the last equality follows by choosing $\hat{\Psi} = (X'X)^{-1}X'Y$, which implies that the cross terms drop out because $X'(Y - X\hat{\Psi}) = X'(Y - X(X'X)^{-1}X'Y) = 0$.

variate normal distribution for Ψ (first line). The matricvariate normal distribution arises when one considers matrices of random variables. If a random matrix Z has a matricvariate distribution, then its vectorization $\text{vec}(Z)$ will have a multivariate normal distribution. A distinguishing feature of the matricvariate normal distribution is the Kronecker structure of its covariance matrix.

A.2. Derivation of posteriors

To simplify notation we define the matrix of OLS residuals cross-products:

$$S = \hat{E}'\hat{E} = (Y - X\hat{\Psi})'(Y - X\hat{\Psi}) = Y'Y - \hat{\Psi}'X'X\hat{\Psi} \quad (58)$$

where the last equality follows from completing the squares and using $X'Y = X'X\hat{\Psi}$. The joint posterior distribution of Ψ and Σ will be proportional to the likelihood in (57) times the prior described in (4):

$$\begin{aligned} P(\Psi, \Sigma | Y) &\propto P(Y | \Psi, \Sigma) P(\Psi, \Sigma) = P(Y | \Psi, \Sigma) P(\Psi | \Sigma) P(\Sigma) \\ &= |\Sigma|^{-0.5M} \exp\{-0.5 \text{tr}[\Sigma^{-1}(\Psi - \hat{\Psi})'X'X(\Psi - \hat{\Psi})]\} \\ &\quad \times |\Sigma|^{-0.5(T-M)} \exp\{-0.5 \text{tr}[\Sigma^{-1}S]\} \times |\Sigma|^{-0.5M} \\ &\quad \times \exp\left\{-0.5 \text{tr}\left[\Sigma^{-1}(\Psi - \Psi_0)' \Omega_0^{-1}(\Psi - \Psi_0)\right]\right\} \\ &\quad \times |\Sigma|^{-0.5(v_0+N+1)} \exp\{-0.5 \text{tr}[\Sigma^{-1}S_0]\}. \end{aligned} \quad (59)$$

In the expression above the last two lines are the density kernels of $\Psi_h | \Sigma \sim N(\Psi_0, \Sigma \otimes \Omega_0)$ and $\Sigma \sim IW(v_0, S_0)$ respectively. Completing the squares appearing in (59) gives:

$$(\Psi - \hat{\Psi})'X'X(\Psi - \hat{\Psi}) = \Psi'X'X\Psi - \Psi'X'X\hat{\Psi} - \hat{\Psi}'X'X\Psi + \hat{\Psi}'X'X\hat{\Psi} \quad (60)$$

$$(\Psi - \Psi_0)\Omega_0^{-1}(\Psi - \Psi_0) = \Psi'\Omega_0^{-1}\Psi - \Psi'\Omega_0^{-1}\Psi_0 - \Psi_0'\Omega_0^{-1}\Psi + \Psi_0'\Omega_0^{-1}\Psi_0 \quad (61)$$

Define:

$$\bar{\Omega} = (\Omega_0^{-1} + X'X)^{-1} \quad (62)$$

$$\bar{\Psi} = \bar{\Omega}(\Omega_0^{-1}\Psi_0 + X'Y) \quad (63)$$

By premultiplying (63) by $\bar{\Omega}^{-1}$ and using $X'Y = X'X\hat{\Psi}$ the latter equation also implies:

$$\bar{\Omega}^{-1}\bar{\Psi} = \Omega_0^{-1}\Psi_0 + X'X\hat{\Psi} \quad (64)$$

Now consider the following quadratic form:

$$\begin{aligned} (\Psi - \bar{\Psi})\bar{\Omega}^{-1}(\Psi - \bar{\Psi}) &= \Psi'\bar{\Omega}^{-1}\Psi - \Psi'\bar{\Omega}^{-1}\bar{\Psi} - \bar{\Psi}'\bar{\Omega}^{-1}\Psi + \bar{\Psi}'\bar{\Omega}^{-1}\bar{\Psi} \\ &= \Psi'(\Omega_0^{-1} + X'X)\Psi - \Psi'(\Omega_0^{-1}\Psi_0 + X'X\hat{\Psi}) \\ &\quad - (\hat{\Psi}'X'X + \Psi_0'\Omega_0^{-1})\Psi + \bar{\Psi}'\bar{\Omega}^{-1}\bar{\Psi} \\ &= \Psi'\Omega_0^{-1}\Psi + \Psi'X'X\Psi - \Psi'\Omega_0^{-1}\Psi_0 - \Psi'X'X\hat{\Psi} \\ &\quad - \hat{\Psi}'X'X\Psi - \Psi_0'\Omega_0^{-1}\Psi + \bar{\Psi}'\bar{\Omega}^{-1}\bar{\Psi} \end{aligned} \quad (65)$$

Contrasting (65) with (60) and (61) we have:

$$(\Psi - \bar{\Psi})\bar{\Omega}^{-1}(\Psi - \bar{\Psi}) + S_\Psi = (\Psi - \hat{\Psi})'X'X(\Psi - \hat{\Psi}) + (\Psi - \Psi_0)\Omega_0^{-1}(\Psi - \Psi_0) \quad (66)$$

where the matrix S_Ψ is:

$$S_\Psi = \hat{\Psi}'X'X\hat{\Psi} + \Psi_0'\Omega_0^{-1}\Psi_0 - \bar{\Psi}'\bar{\Omega}^{-1}\bar{\Psi} \quad (67)$$

and does not depend on Ψ . We can use (66) in (59) to get:

$$\begin{aligned} P(\Psi, \Sigma | Y) &\propto |\Sigma|^{-0.5T} |\Sigma|^{-0.5M} |\Sigma|^{-0.5(v_0+N+1)} \exp\{-0.5 \text{tr}[\Sigma^{-1}S + \Sigma^{-1}S_0]\} \\ &\quad \times \exp\{-0.5 \text{tr}[\Sigma^{-1}((\Psi - \bar{\Psi})\bar{\Omega}^{-1}(\Psi - \bar{\Psi}) + S_\Psi)]\} \\ &= |\Sigma|^{-0.5(v_0+T+N+1)} \exp\{-0.5 \text{tr}[\Sigma^{-1}S + \Sigma^{-1}S_0 + \Sigma^{-1}S_\Psi]\} \\ &\quad \times |\Sigma|^{-0.5M} \exp\{-0.5 \text{tr}[\Sigma^{-1}(\Psi - \bar{\Psi})\bar{\Omega}^{-1}(\Psi - \bar{\Psi})]\} \end{aligned} \quad (68)$$

We further define:

$$\bar{v} = v_0 + T \quad (70)$$

and:

$$\bar{S} = S_0 + S + S_\Psi, \quad (71)$$

which using (58) can be written also as:

$$\bar{S} = S_0 + \hat{E}'\hat{E} + \hat{\Psi}'X'X\hat{\Psi} + \Psi_0'\Omega_0^{-1}\Psi_0 - \bar{\Psi}'\bar{\Omega}^{-1}\bar{\Psi} \quad (72)$$

$$= S_0 + Y'Y + \Psi_0'\Omega_0^{-1}\Psi_0 - \bar{\Psi}'\bar{\Omega}^{-1}\bar{\Psi}. \quad (73)$$

Then we can re-write (69) as:

$$\begin{aligned} P(\Psi, \Sigma | Y) &\propto |\Sigma|^{-0.5(\bar{v}+N+1)} \exp\{-0.5 \text{tr}[\Sigma^{-1}\bar{S}]\} \times |\Sigma|^{-0.5M} \\ &\quad \times \exp\{-0.5 \text{tr}[\Sigma^{-1}(\Psi - \bar{\Psi})\bar{\Omega}^{-1}(\Psi - \bar{\Psi})]\} \end{aligned} \quad (74)$$

Which is the product of a conditional (on Σ) matricvariate normal kernel for Ψ and of an inverse Wishart kernel for Σ :

$$\Psi | \Sigma, Y \sim N(\bar{\Psi}, \Sigma \otimes \bar{\Omega}), \quad \Sigma | Y \sim IW(\bar{v}, \bar{S}), \quad (75)$$

For this case it is also possible to obtain the marginal posterior distribution of Ψ . To do so it is sufficient to invoke theorem A.19 p. 307 in Bauwens et al. (1999), which states that if the random variables Z and V are distributed as $Z|V \sim N$ and $V \sim IW$ respectively, then the marginal distribution of Z is matricvariate t . In this case we have:

$$\Psi | Y \sim MT(\bar{\Psi}, \bar{\Omega}^{-1}, \bar{S}, \bar{v}). \quad (76)$$

Also the matricvariate t arises in the context of matrices of random variables. If a random matrix Z has a matricvariate t distribution, then its vectorisation $\text{vec}(Z)$ will have a multivariate t distribution.

A.3. Derivation of the marginal likelihood

First, we note that the likelihood is (matricvariate) normal:

$$Y | \Psi, \Sigma \sim N(X\Psi, \Sigma \otimes I). \quad (77)$$

To obtain the marginal data density we need to integrate out Ψ and Σ . First, we integrate out Ψ . Using the prior in (4) we can derive the distribution of the expected value $X\Psi$ given Σ :

$$X\Psi | \Sigma \sim N(X\Psi_0, \Sigma \otimes X\Omega_0X'). \quad (78)$$

Noting that $Y = X\Psi + E$, where $E | \Sigma \sim N(0, \Sigma \otimes I)$ is independent from $X\Psi$, we can write $Y | \Sigma$ as:

$$Y | \Sigma = X\Psi | \Sigma + E | \Sigma \sim N(X\Psi_0, \Sigma \otimes (I + X\Omega_0X')). \quad (79)$$

Second, we integrate out the Σ . To do so it is sufficient to note that $Y | \Sigma$ and Σ are a $N - IW$, and invoke theorem A.19 pag. 307 in Bauwens et al. (1999). As a result, the marginal density of Y is a matricvariate t :

$$Y \sim MT(X\Psi_0, (I + X\Omega_0X')^{-1}, S_0, v_0). \quad (80)$$

Using the parameterization of Dickey (1967), the expression for the p.d.f. of Y is:

$$p(Y) = k^{-1} \times |S_0 + (Y - X\Psi_0)'(I + X\Omega_0X')^{-1}(Y - X\Psi_0)|^{-\frac{v_0+T}{2}} \quad (81)$$

with:

$$k = \pi^{\frac{TN}{2}} \times |(I + X\Omega_0 X')^{-1}|^{-\frac{N}{2}} \times |S_0|^{-\frac{v_0}{2}} \times \prod_{i=1}^N \frac{\Gamma\left(\frac{v_0+1-i}{2}\right)}{\Gamma\left(\frac{v_0+T+1-i}{2}\right)}, \quad (82)$$

where $\Gamma(\cdot)$ is the univariate gamma function. In our application, the prior variance matrix Ω_0 is a function of the tightness hyperparameter θ , therefore the marginal likelihood in (81) is computed using $\Omega_0(\theta)$, for a grid of values of θ . In particular we have:

$$\begin{aligned} p_\theta(Y) &= p(Y|\theta) \\ &= \pi^{\frac{TN}{2}} \times |(I + X\Omega_0(\theta)X')^{-1}|^{-\frac{N}{2}} \times |S_0|^{-\frac{v_0}{2}} \times \prod_{i=1}^N \frac{\Gamma\left(\frac{v_0+T+1-i}{2}\right)}{\Gamma\left(\frac{v_0+1-i}{2}\right)} \\ &\quad \times |S_0 + (Y - X\Psi_0)'(I + X\Omega_0(\theta)X')^{-1}(Y - X\Psi_0)|^{-\frac{v_0+T}{2}}. \end{aligned} \quad (83)$$

References

- Almeida, C., Vicente, J., 2008. The role of no-arbitrage on forecasting: lessons from a parametric term structure model. *Journal of Banking and Finance* 32 (12), 2695–2705.
- Ang, A., Piazzesi, M., 2003. A no-arbitrage vector autoregression of term structure dynamics with macroeconomic and latent variables. *Journal of Monetary Economics* 50 (4), 745–787.
- Bai, J., Ng, S., 2006. Confidence intervals for diffusion index forecasts and inference for factor-augmented regressions. *Econometrica* 74 (4), 1133–1150.
- Banbura, M., Giannone, D., Reichlin, L., 2010. Large Bayesian vector autoregressions. *Journal of Applied Econometrics* 25 (1), 71–92.
- Bauwens, L., Lubrano, M., Richard, J.F., 1999. *Bayesian Inference in Dynamic Econometric Models*. Oxford University Press, Oxford.
- Carriero, A., Clark, T., Marcellino, M., 2011a. Bayesian VARs: specification choices and forecasting performance. CEPR discussion papers 8273.
- Carriero, A., Giacomini, R., 2011. How useful are no-arbitrage restrictions for forecasting the term structure of interest rates? *Journal of Econometrics* 164 (1), 21–34.
- Carriero, A., Kapetanios, G., Marcellino, M., 2009. Forecasting exchange rates with a large Bayesian VAR. *International Journal of Forecasting* 25 (2), 400–417.
- Carriero, A., Kapetanios, G., Marcellino, M., 2011b. Forecasting large datasets with Bayesian reduced rank multivariate models. *Journal of Applied Econometrics* 26, 736–761.
- Christensen, J.H.E., Diebold, F.X., Rudebusch, G., 2011. The affine arbitrage-free class of Nelson–Siegel term structure models. *Journal of Econometrics* 164, 4–20.
- Cochrane, J.H., Piazzesi, M., 2005. Bond risk premia. *American Economic Review* 95 (1), 138–160.
- Dai, Q., Singleton, K., 2000. Specification analysis of affine term structure models. *Journal of Finance* 55, 1943–1978.
- De Jong, F., 2000. Time series and cross section information in affine term structure models. *Journal of Business and Economic Statistics* 18, 300–314.
- De Mol, C., Giannone, D., Reichlin, L., 2008. Forecasting using a large number of predictors. Is Bayesian regression a valid alternative to principal components? *Journal of Econometrics* 146 (2), 318–328.
- De Pooter, M., Ravazzolo, F., van Dijk, D., 2007. Predicting the term structure of interest rates incorporating parameter uncertainty, model uncertainty and macroeconomic information. Tinbergen Institute discussion papers 07-028/4.
- Dickey, J.M., 1967. Matricvariate generalizations of the multivariate t distribution and the inverted multivariate t distribution. *Annals of Mathematical Statistics* 38 (2), 511–518.
- Diebold, F.X., Li, C., 2006. Forecasting the term structure of government bond yields. *Journal of Econometrics* 130, 337–364.
- Diebold, F.X., Mariano, R., 1995. Comparing predictive accuracy. *Journal of Business & Economic Statistics* 13 (3), 253–263.
- Doan, T., Litterman, R., Sims, C., 1984. Forecasting and conditional projection using realistic prior distributions. *Econometric Reviews* 3, 1–100.
- Duffee, G., 2002. Term premia and interest rate forecasts in affine models. *Journal of Finance* 57, 405–443.
- Duffee, G., 2011. Forecasting with the term structure: the role of no-arbitrage restrictions. Mimeo, Johns Hopkins University. Available at: <<http://www.econ2.jhu.edu/people/Duffee/>>.
- Duffie, D., Kan, R., 1996. A yield-factor model of interest rates. *Mathematical Finance* 6, 379–406.
- Fama, E., Bliss, R., 1987. The information in long-maturity forward rates. *American Economic Review* 77, 680–692.
- Favero, C.A., Niu, L., Sala, L., 2012. Term structure forecasting: no-arbitrage restrictions versus large information set. *Journal of Forecasting* 31, 124–156.
- Geweke, J., 2005. *Contemporary Bayesian Econometrics and Statistics*. Wiley, New Jersey.
- Giacomini, R., White, H., 2006. Tests of conditional forecast accuracy. *Econometrica* 74, 1545–1578.
- Giannone, D., Lenza, M., Primiceri, G.E., 2012. Prior Selection for Vector Autoregressions. Working Papers ECARES 2012-002, ULB –Université Libre de Bruxelles.
- Gurkaynak, R.S., Sack, B., Wright, J.H., 2007. The US treasury yield curve: 1961 to the present. *Journal of Monetary Economics* 54 (8), 2291–2304.
- Harrison, J.M., Kreps, D.M., 1979. Martingales and arbitrage in multiperiod securities markets. *Journal of Economic Theory* 2, 381–408.
- Ho, M.S., Sørensen, B.E., 1996. Finding cointegration rank in high dimensional systems using the Johansen test: an illustration using data based Monte Carlo simulations. *Review of Economics and Statistics* 78, 726–732.
- Kadiyala, K.R., Karlsson, S., 1997. Numerical methods for estimation and inference in Bayesian VAR-models. *Journal of Applied Econometrics* 12 (2), 99–132.
- Joslin, S., Singleton, Kenneth J., Zhu, H., 2011. A new perspective on gaussian dynamic term structure models. *Review of Financial Studies* 24 (3), 926–970.
- Litterman, R.B., 1986. Forecasting with Bayesian vector autoregressions-five years of experience. *Journal of Business & Economic Statistics* 4, 25–38.
- Marcellino, M., Stock, J.H., Watson, M.W., 2006. A comparison of direct and iterated multistep AR methods for forecasting macroeconomic time series. *Journal of Econometrics* 127 (1–2), 499–526.
- Markowitz, H., 1952. Portfolio selection. *The Journal of Finance* 7 (1), 77–91.
- Moench, E., 2008. Forecasting the yield curve in a data-rich environment: a no-arbitrage factor-augmented VAR approach. *Journal of Econometrics* 146 (1), 26–43.
- Moench, E., 2010. Term structure surprises: the predictive content of curvature, level, and slope. *Journal of Applied Econometrics*. <http://dx.doi.org/10.1002/jae.1220>.
- Nelson, C.R., Siegel, A.F., 1987. Parsimonious modeling of yield curve. *Journal of Business* 60, 473–489.
- Pesaran, M.H., Pick, A., Timmerman, A., 2011. Variable selection and inference for multi-period forecasting problems. *Journal of Econometrics* 164 (1), 173–187.
- Pesaran, M.H., Timmermann, A., 2005. Small sample properties of forecasts from autoregressive models under structural breaks. *Journal of Econometrics* 129 (1–2), 183–217.
- Politis, D.N., Romano, J.P., 1994. The stationary bootstrap. *Journal of the American Statistical Association* 89 (428), 1303–1313.
- Sims, C., Zha, T., 1998. Bayesian methods for dynamic multivariate models. *International Economic Review* 39 (4), 949–968.
- Stock, J.H., Watson, M.W., 2002a. Macroeconomic forecasting using diffusion indexes. *Journal of Business and Economic Statistics* 20, 147–162.
- Stock, J.H., Watson, M.W., 2002b. Forecasting using principal components from a large number of predictors. *Journal of the American Statistical Association* 97, 1167–1179.
- Svensson, L.E.O., 1994. Estimating and interpreting forward interest rates: Sweden 1992–4. CEPR discussion papers 1051.
- Tiao, G., Zellner, A., 1964. On the Bayesian estimation of multivariate regression. *Journal of the Royal Statistical Society Series B* 26, 277–285.
- Zellner, A., 1973. *An Introduction to Bayesian Inference in Econometrics*. Wiley, New Jersey.